

Analysing the uncertainties of reanalysis data used for wind resource assessment: A critical review

G. Gualtieri

National Research Council–Institute for BioEconomy (CNR-IBE), Via Caproni 8, 50145, Firenze, Italy

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ABSTRACT

This review critically updates the state of the art on the uncertainties associated with reanalysis data directly used for wind resource assessment. Performances were analysed from 15 (9 global and 6 regional) reanalysis products applied on 322 very heterogeneous locations worldwide. Reanalysis scores were assessed by location type (offshore, coastal, inland and mountainous) and height above the ground (10–300 m).

The results confirmed that reanalyses can predict wind resource pattern over time, while they are not similarly able over space. Reanalysis data (particularly by ERA5) are sufficiently reliable on offshore and flat onshore locations. Uncertainties are greater on mountainous and coastal sites, where wind speed and energy yield are significantly under- and over-estimated, respectively. On these locations, the use of higher-resolution regional products should be preferred over global datasets as they are better suited to resolving detailed terrain features as well as roughness and topographical discontinuities. Comparison between older and newer reanalysis versions revealed that the latter do not always outperform the former. In many cases, however, improvements by the newer products were found – rather than in accuracy itself – in pushing a given accuracy towards higher elevations, thus more suitably meeting the needs of modern wind power industry. In perspective, it is desirable that advances in forecasting models will enable current problems in assimilating surface land wind observations by global reanalyses to be overcome. Refinements in global datasets could also positively reflect on regional products.

1. Introduction

The highly fluctuating nature of wind requires a probabilistic approach to perform a site's wind resource assessment. Wind data should be collected over a long time period (spanning several years) [1] at high spatial (a few kilometres) and temporal (hours) scales [2]. Ideally, wind data should be measured at a suitable WT hub height, that is nowadays between 80 and 100 m onshore [3], and between 80 and 120 m offshore [4,5]. However, in the vast majority of real-world cases wind speed is measured at about 10 m, as is typically the case for weather stations (onshore), and moored buoys (offshore). These surface wind measurements must then be extrapolated to WT hub height using various extrapolation models, which inevitably introduce some approximations [6]. Instead, met masts should probably be regarded as the most reliable source of wind resource information. A recently developed Tall Tower Database, gathering met data from 222 existing tall towers around the world [7], may be an effective solution to this issue. However, met masts may be affected by limited temporal coverage, relocation, device updates, and changes in local topography [8].

A valid (and relatively cheap) way to tackle this problem is to use atmospheric reanalysis data. Global reanalysis products integrate three components (a global forecasting model, input observations and an assimilation scheme) in order to accurately simulate past states of the atmosphere [9]. Observations are input every 6–12 h through an unchanging (“frozen”) combination of forecasting model and assimilation scheme, so that a dynamically-consistent state of the atmosphere is estimated at each time step. Thus, while data assimilation systems used to produce operational analyses are continuously updated to improve NWP, reanalyses are produced by using frozen data assimilation system for the whole period [9]. Combining assimilation of historic observations from various monitors (ground stations, met towers, buoys, ships, etc.) with an atmospheric circulation model to infer the state of the global weather system, reanalyses produce fully-consistent datasets for a very long time period (typically more than 30 years) [10]. Unlike observed data, only retrieving point-based (and often sparsely spread) information, reanalyses provide meteorological data on a global regular grid, thus representative for the entire grid cell [11]. Unlike wind atlases, which only retrieve summarized statistics such as monthly or annual averages, reanalyses provide actual 4D information. Reanalysis

E-mail address: giovanni.gualtieri@ibe.cnr.it.

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Nomenclature**Abbreviations**

3D-Var	Three-dimensional variational data assimilation
4D-Var	Four-dimensional variational data assimilation
ANN	Artificial neural network
CFSR	Climate Forecast System Reanalysis
CFSv2	Climate Forecast System Reanalysis version 2
CRAN	Comprehensive R Archive Network
COSMO	Consortium for Small-scale Modelling
DWD	Deutscher Wetterdienst
ECMWF	European Centre for Medium-Range Weather Forecasts
GWA	Global Wind Atlas
IFS	Integrated Forecasting System
IQR	Interquartile range (Q3–Q1)
LIDAR	Laser Detection and Ranging
MERA	Met Éireann ReAnalysis
MERRA	Modern-Era Retrospective Analysis for Research and Application
MERRA2	Second Modern-Era Retrospective analysis for Research and Applications
MM5	Fifth-Generation NCAR/Penn State Mesoscale Model
NARR	North American Regional Reanalysis
NASA	National Aeronautics and Space Administration
NCAR	National Center for Atmospheric Research
NCEP	National Centers for Environmental Prediction
NWP	Numerical weather prediction
Q1,Q2,Q3	1st, 2nd, and 3rd quartile
R1	NCEP/NCAR Reanalysis 1
R2	NCEP/NCAR Reanalysis 2
RAMS	Regional Atmospheric Modeling System

SMHI	Swedish Meteorological and Hydrological Institute
UERRA	Uncertainties in Ensembles of Regional ReAnalysis
UM	Unified Model
WAsP	Wind Atlas Analysis and Application Program
WRF	Weather Research and Forecasting
WT	Wind turbine

Variables

AEY	annual energy yield [GWh/y] = $P_{ave} * 8760 * (1 - F_{tot})$, where P_{ave} is site's average WT power output and F_{tot} is WT total power losses
CF	capacity factor [%] = AEY/E_r , where E_r is the energy produced by the WT if operated at its rated power across the same period

Statistical skill scores

N	number of predictions/observations
P_i	predictions [m/s or GWh/y]
$\bar{P}$	mean predictions [m/s or GWh/y] = $\frac{1}{N} \sum_{i=1}^N P_i$
O_i	observations [m/s or GWh/y]
$\bar{O}$	mean observations [m/s or GWh/y] = $\frac{1}{N} \sum_{i=1}^N O_i$
MB	wind speed mean bias [m/s] = $\frac{1}{N} \sum_{i=1}^N (P_i - O_i)$
$RMSE$	wind speed root mean square error [m/s] = $\sqrt{\frac{1}{N} \sum_{i=1}^N (P_i - O_i)^2}$
R	wind speed correlation coefficient = $\frac{\sum_{i=1}^N (P_i - \bar{P})(O_i - \bar{O})}{\sqrt{\sum_{i=1}^N (P_i - \bar{P})^2} \sqrt{\sum_{i=1}^N (O_i - \bar{O})^2}}$
PE	AEY percent error [%] = $100 * (\bar{P} - \bar{O}) / \bar{O}$

data are massively used within various scientific disciplines, mainly climatological studies. In recent years, they have been rapidly gaining popularity in the wind energy sector as they offer a number of advantages, including: (i) public availability, as they can be freely downloaded; (ii) global (worldwide) coverage, so that the wind resource can also be assessed at remote land locations or sites far offshore; (iii) long-term time series, thus allowing not only to tackle possible year-to-year wind speed anomalies that increase the risks of a wind farm project, but also to predict changes in wind farm performance over their lifetime [12]; (iv) spatial and temporal completeness, thus addressing spatial discontinuities and temporal gaps. A clear limitation of reanalysis data is their coarse spatial resolution (from tens to hundreds of kilometres), which prevents them from fully resolving not only complex topographic features of a specific area (e.g. Ref. [13]), but also marked land use heterogeneity, and thus highly changing roughness conditions (e.g. Ref. [14]). To address this issue, the prevailing approach is to use NWP models to downscale reanalysis data into smaller and more detailed spatial grids [10]. Previous wind energy applications include, for example, the RAMS model [15] forced by the R2 reanalysis [16], MM5 [17] forced by NARR [18], ALADIN [19] forced by ERA40 [20], as well as WRF [21] forced by a number of different reanalyses such as R1 [22], R2 [23], CFSR [24], MERRA2 [25], ERA40 [22], ERA-Interim [26], and ERA5 [27]. However, this downscaling process is costly in terms of time, expertise, hardware/software requirements, computation resources and data demands [10]. Unlike global reanalysis products (covering the whole world), several regional reanalysis products (limited to specific areas such as continents, subcontinents or even countries) have been developed to provide higher spatial resolution. They are refinements of a global reanalysis performed by means of dynamic downscaling and data assimilation techniques [12], so that they can be considered as a valid tradeoff between coarser global reanalyses and finer NWP models. However, their shorter timespan and limited

spatial coverage are the price to pay for higher spatial detail than global reanalyses [28]. Furthermore, they depend on the global reanalysis used as boundary conditions, so can be affected by any potential issue (including phasing out) of global reanalyses [11].

In the last few years, remarkable improvements in reanalysis datasets have encouraged their use as a direct input for wind resource assessment (e.g. Refs. [4,29–37]). The purpose of this paper is therefore to achieve a critical understanding of the advances in (both global and regional) reanalysis datasets directly used for wind energy applications. This was done by reviewing the most prominent research studies that assessed reanalysis accuracy while meeting the following conditions: (i) a comparison is made with individual in-situ observations; (ii) a minimum period of ~1 year with a time resolution not exceeding 6 h is considered; (iii) at least one variable between wind speed and energy yield is addressed; (iv) at least one metric is calculated among bias, root mean square error and correlation coefficient (for wind speed), and percent error (for energy yield). Indeed, several studies in the literature have already focused on assessing reanalysis uncertainties on a global [38–40], continental [41], country [10,37,42,43] and regional [3, 44–46] scale. However, with a few exceptions (e.g. Refs. [2,27]), these analyses were conducted on a purely geographical basis and scores aggregated by region or country. The main contribution of this review is instead to critically update the current state of the art according to a more thorough and comprehensive overview by analyzing applications: (i) of the broadest spectrum of existing reanalysis datasets (both global and regional); (ii) performed worldwide across the longest time series; (iii) against the most heterogeneous locations, after having classified the sites into four types based on environment, altitude and topography; (iv) as a function of height above the ground.

Table 1
Characteristics of the selected reanalysis products^{a,b}.

Product	Institution	Spatial coverage	Release year	Temporal coverage	Time res. (h)	Spatial resolution (Lat-Lon at the equator)	Levels	Wind speed height(s) (m a.g. l.)	Ref.
Global reanalysis									
R1	NCEP/NCAR	World	1995	1948–present	6	$1.875^{\circ} \times 2.0^{\circ}$ (~210 km)	28 up to 3 hPa	10	[47]
R2	NCEP/NCAR	World	2000	1979–present	6	$1.875^{\circ} \times 2.0^{\circ}$ (~210 km)	28 up to 3 hPa	10	[48]
ERA40	ECMWF	World	2003	1957–2002	6	$1.125^{\circ} \times 1.125^{\circ}$ (~125 km)	60 up to 0.1 hPa	10	[53]
ERA-Interim	ECMWF	World	2008	1979–2019	6	$0.75^{\circ} \times 0.75^{\circ}$ (~79 km)	60 up to 0.1 hPa	10	[54]
CFSR	NCEP	World	2009	1979–2010	1	$0.3125^{\circ} \times 0.3125^{\circ}$ (~38 km)	64 up to 0.266 hPa	10	[49]
MERRA	NASA	World	2009	1979–2016	1	$0.5^{\circ} \times 0.667^{\circ}$ (~55 km)	72 up to 0.01 hPa	10, 50	[51]
CFSv2	NCEP	World	2011	2011–present	1	$0.205^{\circ} \times 0.205^{\circ}$ (~23 km)	64 up to 0.266 hPa	10	[50]
MERRA2	NASA	World	2015	1980–present	1	$0.5^{\circ} \times 0.625^{\circ}$ (~55 km)	72 up to 0.01 hPa	10, 50	[52]
ERA5	ECMWF	World	2019	1950–present	1	$0.25^{\circ} \times 0.25^{\circ}$ (~31 km)	137 up to 0.01 hPa	10, 100	[55]
Regional reanalysis									
NARR	NCEP	North America	2006	1979–2003	3	32 km	45	10, 30	[57]
COSMO-REA6	COSMO	Europe	2015	1995–2019	1	$0.055^{\circ} \times 0.055^{\circ}$ (~6 km)	40 up to 40 hPa	10, 20, 40, 60, 80, 100	[58]
COSMO-REA2	COSMO	Central Europe	2017	2007–2013	1	$0.018^{\circ} \times 0.018^{\circ}$ (~2 km)	50	10, 20, 40, 60, 80, 100	[59]
HARMONIEv1	SMHI	Europe	2017	1961–2015	6	$0.11^{\circ} \times 0.11^{\circ}$ (~12 km)	65 up to 10 hPa	10, 15, 30, 50, 75, 100, 150, 200, 250, 300, 400, 500	[60]
UM	Met Office	Europe	2017	1979–2016	6	$0.11^{\circ} \times 0.11^{\circ}$ (~12 km)	65 up to 10 hPa	10, 15, 30, 50, 75, 100, 150, 200, 250, 300, 400, 500	[61]
MÉRA	Met Éireann	Ireland and UK	2017	1981–2015	3	2.5 km	65 up to 10 hPa	10, 30, 50, 60, 70, 80, 90, 100, 125, 150, 200, 300, 400	[62]

^a Global and regional products are ordered by release year.

^b Implemented data assimilation system: 3D-Var for R1, R2, ERA40, CFSR, CFSv2, MERRA2, NARR, HARMONIEv1, and MÉRA; 4D-Var for ERA-Interim, ERA5 and UM; Nudging for COSMO-REA2 and COSMO-REA6.

2. Reanalysis products

Only reanalysis products for which articles meeting the current selection criteria have been found in the literature (see later) will be described and analysed.

2.1. Global reanalyses

In 1995, the NCEP/NCAR released the NCEP/NCAR Reanalysis 1 (R1, [47]) to cover a 40-year period (1957–1996). This period was then extended to span from 1948 to the present (Table 1). R1 provides 6-h estimations (including 10-m wind speed) gridded according to a $1.875^{\circ} \times 2.0^{\circ}$ (~210 km) spatial resolution over 28 vertical levels. Wind data are assimilated from radiosondes, aircraft, buoys, ships, while surface winds are neither assimilated from land stations (onshore) nor from satellites (offshore). In 2000, the NCEP/NCAR Reanalysis 2 (R2, [48]) was released. R2 corrected various processing errors encountered in its predecessor (R1) and improved data assimilation, while maintaining the same spatial, vertical and temporal resolutions. Although R2 assimilates a limited amount of satellite data, it assimilates ocean surface winds from satellites. Both R1 and R2 products implement CDAS and are still operational at NCEP/NCAR.

The NCEP CFSR is available from 1979 to 2011 [49]. CFSR spatial resolution is $0.3125^{\circ} \times 0.3125^{\circ}$ (~38 km), implementing 64 vertical levels up to 0.266 hPa. Wind data at 10 m with a 1-h time resolution are provided. CFSR is superior to previous NCEP reanalyses (R1 and R2) not only due to higher spatial and temporal resolution, but also improved forecasting model and assimilation scheme, as well as coupled atmosphere-land-ocean-sea ice system. Wind data are assimilated from

radiosondes, aircraft, ships, buoys and satellites. NCEP upgraded CFSR to version 2 (CFSv2) in March 2011 through finer spatial resolution ($0.205^{\circ} \times 0.205^{\circ}$) and massive use of satellite data. Data are available as 1-h averages from 2011 to the present [50]. Both CFSR and CFSv2 products implement GDAS and are still operational at NCEP/NCAR.

MERRA is a global reanalysis released by NASA in 2009 [51]. MERRA spatial resolution is $0.5^{\circ} \times 0.667^{\circ}$ (~55 km), while atmospheric vertical structure is described using 72 levels up to 0.01 hPa. Data are available as 1-h averages from 1979 to 2016. Wind data are collected from balloons, radiosondes, ships, buoys, aircraft and satellites. MERRA implements the GEOS-5 v. 5.2.0 model. Notably, MERRA was the first reanalysis dataset providing wind speed not only at 10 m (as earlier products did), but also at 50 m. MERRA was terminated in March 2016. In 2015, NASA updated MERRA with MERRA2 [52]. Compared to its predecessor, MERRA2 implements newer advances in atmospheric modelling (GEOS-5 v. 5.12.4) and data assimilation, while also incorporating estimations from newer satellite instruments. MERRA2 has the same temporal resolution and coverage as MERRA, while spatial resolution is improved in longitude (0.625° vs. 0.667°). MERRA2 is still operational at NCEP/NCAR.

ERA40 is a global reanalysis product developed in 2003 by the ECMWF. Data are available from Sept. 1957 through Aug. 2002, as 6-h averages with a spatial resolution of $1.125^{\circ} \times 1.125^{\circ}$ (~125 km) over 60 vertical levels up to 0.1 hPa [53]. Wind data are assimilated by radiosondes, balloons, aircraft, buoys, and (since 1979) satellites. ERA40 implements the Cy23r4 IFS. In 2008, ERA40 was replaced by ERA-Interim, which implemented a largely improved atmospheric model (Cy31r2) and assimilation scheme and was terminated at the end of Aug. 2019. ERA-Interim data span from Jan. 1979 to Aug. 2019 and

include 6-h estimates spaced at a resolution of $0.75^\circ \times 0.75^\circ$ (~ 79 km) over 60 vertical levels [54]. Released in 2019, ERA5 is currently the latest version of the ECMWF global reanalysis products. It provides 1-h estimates of several variables spaced at a resolution of $0.25^\circ \times 0.25^\circ$ (~ 31 km) using 137 vertical levels up to 0.01 hPa. Data are available from 1950 to the present and include wind speed not only at 10 m (like its predecessors), but also at 100 m, currently the highest elevation among all global reanalyses. ERA5 implements the Cy41r2 IFS [55].

Other relevant global reanalysis products – not covered in this review – are available, such as, e.g., the Japanese 55-year Reanalysis (JRA55, [56]), which provides 6-h estimations from 1958 to the present through a $0.5625^\circ \times 0.5625^\circ$ (~ 55 km) spatial resolution over 60 vertical levels.

2.2. Regional reanalyses

NARR was developed by NCEP as an extension of R1 to run on the North American domain. It covers a 25-y period (1979–2003) through 3-h data, with a 32-km spatial resolution and 45 vertical levels. Wind data are available at 10 and 30 m. NARR implements the NCEP Eta model, while lateral boundary conditions are provided by R2 [57]. Improvements in the model/assimilation schemes have resulted in a more accurate dataset than previous global reanalysis datasets, particularly in wind speed simulation.

Several regional reanalysis products have recently been developed to cover Europe (or a part of it) within the framework of the UERRA project. They include COSMO-REA6, COSMO-REA2, HARMONIEv1 and UM.

Within the DWD Hans Ertel Centre for Weather Research, the University of Bonn produced the deterministic regional reanalysis COSMO-REA6, covering Europe with a $0.055^\circ \times 0.055^\circ$ (~ 6 km) spatial resolution and using 40 vertical levels ranging up to 40 hPa [58]. In addition to radiosondes, aircraft and ships, wind data are assimilated from land surface (10 m) and upper (up to 100 m) stations, while no satellite data are incorporated. Boundary conditions are provided by the ERA-Interim global reanalysis, so that COSMO-REA6 was discontinued when it was ERA-Interim (i.e. in Aug. 2019). The COSMO model is used to perform the dynamic downscaling. Wind data are provided as 1-h averages at various heights. COSMO-REA2 is a regional reanalysis product covering central Europe (i.e. Germany and parts of the neighbouring countries) with a $0.018^\circ \times 0.018^\circ$ (~ 2 km) spatial resolution over 50 vertical levels. Lateral boundary conditions are provided by COSMO-REA6. COSMO-REA2 assimilates the same wind observations as COSMO-REA6. At present, data cover the period 2007–2013 and are available as 1-h averages on several vertical levels [59].

HARMONIEv1 is a regional reanalysis product developed by the SMHI covering Europe with a $0.11^\circ \times 0.11^\circ$ (~ 12 km) spatial resolution using 65 vertical levels up to 10 hPa. Wind data are assimilated from radiosondes, ships, buoys and aircraft, but not from satellites. Boundary conditions are provided by ERA40 (until 1978) and ERA-interim (1979 onwards), while the HARMONIE-AROME model is used to perform the downscaling. Data are available from 1961 until 2015 as 6-h averages over 12 heights (from 10 to 500 m) [60].

UM is a deterministic reanalysis product developed by the UK Met Office covering Europe from 1979 to 2016. Data are available with a 6-h time step at a resolution of $0.11^\circ \times 0.11^\circ$ (~ 12 km). As with HARMONIEv1, wind data are available at 12 heights (from 10 to 500 m). Satellite, conventional surface and upper air wind data are assimilated. UM is forced by ERA-Interim global reanalysis, whose boundary data are downscaled using the UM model [61].

MERA was developed by The Irish Meteorological Service in 2017. It covers Ireland and the UK using a 2.5-km spatial resolution over 65 vertical levels up to 10 hPa. MERA was forced by ERA-Interim global reanalysis and implements the HARMONIE-AROME model to perform the downscaling. Data are available as 3-h averages at various heights over a 35-y period (1981–2015) [62].

Other relevant regional reanalysis products have been developed

Table 2

Criteria set for paper selection.

Item	Criteria	Comments
Target		
Product	Global or regional reanalysis	Products such as, e.g., analysis, operational forecast, or NWP models not considered
Applications	Direct use of reanalysis	Model coupling/downscaling, statistical (e.g. ANN-based) manipulation, or bias correction not considered
Reference for comparison		
Type of data	Measured	Types include, e.g., met stations, met masts, vertical profilers, buoys, etc.
Spatial aggregation	- Point - Wind farm	Aggregation to clusters of wind farms, areas, regions or countries not considered
Comparison temporal details		
Time span	$\geq \sim 1$ year	At least ~ 1 full year shall be available
Time resolution	≤ 6 h	
Spatial and vertical adjustments of either observations or estimations		
Spatial interpolation	No filter	
Vertical interpolation or extrapolation	No filter	
Target of computations		
Variables/parameters	- Mean wind speed - AEY or CF	At least one of these two variables should be available
Time aggregation	Overall	Sub-yearly analysis (e.g. seasonal, monthly, etc.) not considered
Statistical skill scores		
Mean wind speed	- MB (m/s) - RMSE (m/s) - R	At least one of these four scores should be available
AEY or CF	PE (%)	

worldwide, including, e.g., the Arctic System Reanalysis (ASR v2, [63]) or the Bureau of Meteorology Atmospheric high-resolution Regional Reanalysis for Australia (BARRA v1, [64]).

2.3. Data assimilation system

The implemented data assimilation system is a further feature characterising the reanalysis products. Variational assimilation is the most widely used of such systems, aimed at minimizing the sum of deviations of analysed values from the observed and background forecast values [53]. Three-dimensional (3D-Var) and four-dimensional (4D-Var) variational analysis systems are generally implemented: in 3D-Var the analysis increment is referred to the middle of the assimilation time window, while in 4D-Var the analysis increment evolves in time within each window [54]. The latter is a major step forward from the former [54,55], albeit requiring greater computer resources [56]. Nudging is a simpler yet less accurate method than both 3D-Var and 4D-Var, aimed at dynamically adjusting model's forecast toward the observations by introducing a relaxation term [58]. Among all considered reanalysis products, only ERA-Interim, ERA5 and UM implement the 4D-Var assimilation system, while 3D-Var is implemented by all other products except the COSMO datasets, which use the nudging technique (Table 1).

3. Methods

3.1. Literature analysis and paper selection

The flowchart of paper selection procedure is presented in Fig. S1.

Table 3
Summary of the studies reviewed^a.

Publication		Reanalysis products	Locations			Observations				Calculated variables	
Ref.	Year		No.	Type	Countries	Monitor	Height(s) (m a.g.l.)	Time span (years)	Time bin (h)	Wind speed	AEY
[1]	2021	MERRA2	1	Coastal	India	Met mast	50	1	1	X	
[70]	2021	ERA5	57	Offshore	UK, Denmark, Belgium	Wind farm	58–165	5	1		X
[14]	2021	ERA5	6	Offshore, Coastal, Inland, Mountainous	Australia, Germany, Iran, Netherlands, South Africa, USA	Met mast	40–100	1–2	1	X	X
[29]	2021	ERA5	13	Coastal	Colombia	Met station	10	10	1	X	
[30]	2021	ERA-Interim	8	Coastal, Inland, Mountainous	India	Met station	10	8	1	X	
[71]	2021	ERA5, MERA, MERRA2	7	Coastal, Inland, Mountainous	Ireland	Met station	10	25	1	X	
[72]	2021	ERA5	8	Offshore	USA	Buoy	10	1.5	1	X	
[73]	2021	ERA5	15	Offshore	Brazil	Buoy	10	1–17	1	X	
[74]	2021	ERA5, MERRA2, COSMO-REA2	8	Coastal, Inland, Mountainous	Denmark, Germany	Met mast, LIDAR	60–200	1–3	1	X	
[75]	2021	ERA5	2	Coastal	Romania	Wind farm	39.7–42.6	2.5	1	X	
[76]	2020	MERRA2	1	Offshore	USA	LIDAR	120	1.1	1	X	
[45]	2020	ERA5, MERRA2, NARR	2	Offshore	USA	Buoy	90	30	1	X	
[77]	2020	MERRA2	25	Coastal, Inland, Mountainous	Mexico	Met station	80	1–2	1	X	X
[78]	2020	ERA5, HARMONIEv1, MERRA2	4	Offshore	Denmark, Finland, Germany, Sweden	LIDAR	150, 300	1–3.1	1	X	
[79]	2020	CFSR, ERA-Interim	1	Coastal	Turkey	Met station	10	~1	1	X	
[28]	2020	COSMO-REA2, COSMO-REA6, MERRA2	3	Mountainous	Switzerland	Wind farm	78–100	2.8–6.25	1	X	X
[80]	2020	CFSv2, ERA5, MERRA2	5	Offshore	Brazil	Buoy	10	1.25–7.75	1	X	
[81]	2020	COSMO-REA2, COSMO-REA6, ERA-Interim, MERRA2	4	Coastal, Inland	Germany, Netherlands	Met mast	10–250	5.4–7	3	X	X
[13]	2020	COSMO-REA6, ERA5, MERRA2	8	Coastal, Inland, Mountainous	France	Met mast, LIDAR	55–100	1–3	1	X	
[82]	2019	MERRA2	1	Inland	Cina	Wind farm	50	1	1	X	
[83]	2019	COSMO-REA6, ERA-Interim, HARMONIEv1, UM	4	Offshore, Inland	Germany, Netherlands	Met mast	10–140	5	6	X	
[84]	2019	COSMO-REA2, COSMO-REA6	14	Inland, Mountainous	Czech Republic	Met station	10	3–5	1	X	
[85]	2019	ERA5, MERRA	14	Coastal, Inland, Mountainous	Poland	Met station	100	2	1		X
[86]	2018	CFSR	3	Coastal	Malaysia	Met mast	10	1	1	X	
[87]	2018	MERRA	9	Coastal, Inland, Mountainous	Argentina	Met station	10	2	1	X	
[88]	2017	CFSR, MERRA	8	Offshore	Portugal, Spain	Buoy	10	1	1	X	
[89]	2017	MERRA	11	Coastal, Inland	Ireland	Met station	10	1	1	X	
[90]	2017	ERA5 ^b	5	Coastal, Inland	Latvia	Met station	10	2	2	X	
[91]	2015	MERRA	7	Inland	Germany	Wind farm	79–140	1.5–2.5	1	X	
[92]	2015	NARR	6	Offshore	USA	Buoy	10	5	3	X	
[93]	2015	CFSR	12	Offshore	UK	Buoy, Lightship	10	30	1	X	
[94]	2014	CFSR, ERA-Interim, MERRA, R2	5	Offshore	Spain	Buoy	10	1	1, 6	X	
[41]	2014	CFSR, ERA-Interim	1	Offshore	Peru	Buoy	10	40	3	X	
[95]	2013	R2	11	Inland	Poland	Met mast	61, 83	3–5	6	X	
[96]	2013	CFSR	23	Offshore	USA	Buoy	10	30	1	X	
[97]	2011	CFSR, MERRA, R2	24	Inland	Sweden	Met mast	42, 50	30	1, 6	X	
[98]	2009	ERA-Interim	1	Inland	Hungary	Wind turbine	65	2	6	X	
[99]	2009	R1	5	Offshore	UK, Germany	Buoy, Lightship	10	1	1	X	
[100]	2006	ERA40, R2	5	Offshore	UK	Lighthouse, met mast	10	2–7	6	X	
[101]	2004	ERA40, R1	2	Offshore	Peru	Buoy	10	1	6	X	

^a Studies are in descending order by publication year.

^b The first batch of ERA5 (released in July 2017) covering the period 2010–2016 with a 2-h time resolution was used.

The widely-used Scopus indexing database has been used to retrieve the studies selected in this review. Scopus has been preferred over Web of Science, another commonly used database, due to its broader coverage. The query equation applied is the following:

(TITLE-ABS-KEY (“reanalysis”) AND TITLE-ABS-KEY (“wind speed”) AND TITLE-ABS-KEY (“wind resource” OR “wind power” OR “wind

energy”)) AND (LIMIT-TO (PUBSTAGE, “final”)) AND (LIMIT-TO (DOCTYPE, “ar”) OR LIMIT-TO (DOCTYPE, “cp”)) AND (LIMIT-TO (LANGUAGE, “English”)) AND (LIMIT-TO (SUBJAREA, “ENER”) OR LIMIT-TO (SUBJAREA, “EART”) OR LIMIT-TO (SUBJAREA, “ENGI”) OR LIMIT-TO (SUBJAREA, “ENVI”)).

The entire body of articles has later been subjected to careful analysis

aimed at selecting only those studies exactly matching the scope of the review. This has required establishing a set of specific criteria to be met. As detailed in Table 2, these criteria focus on those articles solely dealing with (global or regional) reanalysis products, thus ignoring, e.g., analysis, forecasting, or NWP models. Only applications that use reanalysis data as direct input have been considered, so that, e.g., model coupling/downscaling (use of reanalysis as initial and/or boundary conditions), statistical manipulation, or bias correction have been discarded. As a reference for comparison, only observed data (e.g. from met stations, met masts, vertical profilers, buoys, lightships) have been considered, being also collected from point monitors or at most from wind farms, thus excluding spatial aggregation to clusters of wind farms, regions, or even countries. A minimum timespan of ~ 1 year [1] with a time resolution not higher than 6 h has been considered.

To increase their accuracy – without using model downscaling –, reanalysis data are generally spatially interpolated from original grid points to the target location(s) using methods such as nearest neighbour, bilinear interpolation, inverse distance weighting, etc. [65]. Likewise, to match their height for comparison, either reanalysis or observation data may be vertically interpolated or extrapolated using models such as the logarithmic or power law (e.g. Ref. [45]). As for these adjustment procedures performed in the literature, no filter has been applied here.

The review is limited to those articles that validate the reanalysis products through at least one variable between mean wind speed and AEY (or alternatively CF), all calculated as overall (i.e. not sub-yearly) values. At least one of the following metrics should be available: MB, RMSE, or R for wind speed; plus PE for AEY. R was not considered for the latter due to the very limited availability of applications in the literature. For convenience, when MB is not available, this value is approximated as the difference between mean predictions and mean observations. The equations for AEY and CF, as well as of the four statistical scores are provided in the Nomenclature section.

3.2. Site classification

In order to discriminate reanalysis scores by site characteristics, locations have been classified based on their environment, altitude and topography. Consistently with other Authors (e.g. Refs. [2,42]), the following classification is set: (i) offshore sites; (ii) coastal sites, i.e. located within ~ 10 – 15 km of the coastline; (iii) inland sites, i.e. basically flat or at most rolling sites with elevation lower than 500 m; (iv) mountainous sites, i.e. sites above 500 m with possibly complex terrain and/or land use (thus possibly including forested sites).

3.3. Computation of reanalysis performance

All computations have been performed using the CRAN “R-stat” environment v. 4.0.3 [66]. Performances of all reanalysis products in simulating wind speed over each location type have been assessed by calculating the frequency distribution of MB, RMSE and R with respect to the observations. Frequency distributions have been analysed through boxplots (Fig. S2) using the “boxplot” function implemented in the R Graphics Package [67]. Capability of reanalysis products in estimating AEY on each location type has been assessed by calculating the PE with respect to the values observed at each height. This has been done using both the “boxplot” function and the “scatterplot” function of the “car” R Package [68]. All statistics have been calculated at 95% confidence level, which required applying a bootstrap resampling to each dataset and then calculating the confidence interval. The “boot” R package [69] has been used to perform this procedure, by using the “boot” function for bootstrap resampling, and the “boot.ci” function for calculating the lower and upper boundaries of the confidence interval.

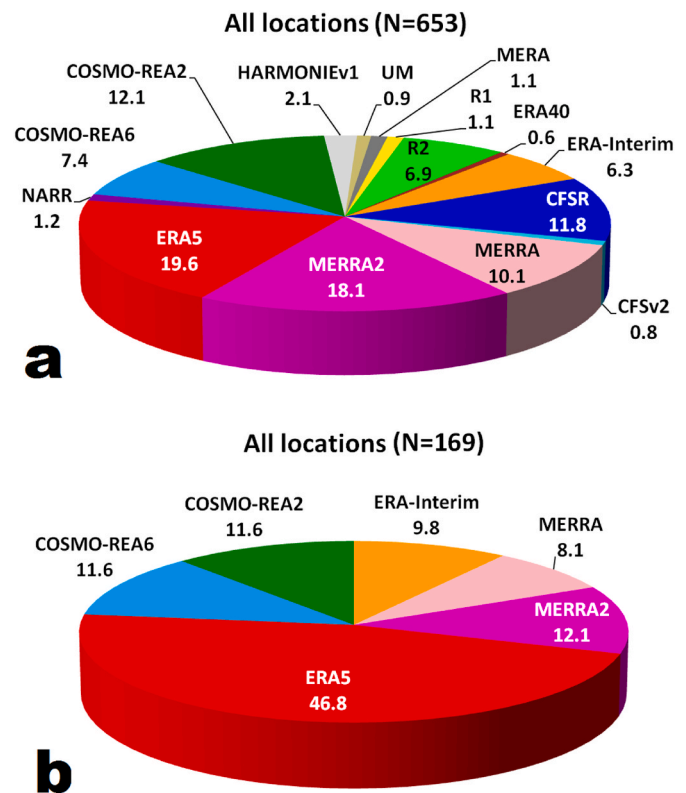


Fig. 1. Pie chart and share (%) of the overall sample (N) of all reanalysis products calculating (a) wind speed and (b) AEY.

4. Results and discussion

4.1. Overview on the selected studies

A total of 264 research articles were initially retrieved after applying the query equation on Scopus (see section 3.1) on Nov. 16, 2021 (Fig. S1). The second stage of the analysis, conducted by applying the filtering criteria set out in Table 2, finally returned a total of 40 articles, which constitute the definitive body of literature reviewed. Overall, performances of 15 (9 global and 6 regional) reanalysis products were evaluated (Table 1). Details of the studies reviewed are presented in Table 3.

Overall, studies range from 2004 to 2021. A significant boost has been received following the release of products such as MERRA (2009), MERRA2 (2015) and ERA5 (2019), particularly as they are the only global reanalyses that provide wind data at WT hub heights (Table 1), trying to somehow meet the needs of the wind power industry [12]. More than 50% of studies have been published since 2019, and more than 75% since 2015. A total of 322 different locations worldwide were analysed: all continents (except Antarctica) were involved, mostly Europe (55%) and America (32%), for a total of 30 countries. In the vast majority of cases, reanalysis products were assessed in calculating wind speed (653 applications), while in fewer cases in calculating AEY (169 applications), the latter not earlier than 2019. With few exceptions, mostly applying to older studies, comparison was performed based on 1-h time bin.

In Fig. S3, the breakdown by location type of reanalysis applications in calculating wind speed (Fig. S3a) and AEY (Fig. S3b) is presented. The preferred locations were generally inland, likely due to availability of reference monitors such as met stations (25%) or met masts (25%), while the remarkable availability of ocean buoys (30%) fostered several offshore studies. Applications on coastal and mountainous locations were fewer. Fig. S4 also allows applications to be discriminate by height

Table 4

Statistics by location type of wind speed scores achieved from all reanalysis products^a.

Locations	Valid data sample (%) ^b	Range	Mean	Std. dev.	Median
Offshore					
Height (m)	174 (27.3)	10–300	48.0	80.5	10.0
MB (m/s)	126 (29.2)	−1.63; +1.54	−0.14	0.62	−0.10
RMSE (m/s)	108 (33.9)	1.07–5.84	2.04	0.89	1.86
R	138 (23.6)	0.572–0.956	0.865	0.081	0.891
Coastal					
Height (m)	88 (13.8)	10–250	59.9	66.2	24.9
MB (m/s)	71 (16.5)	−1.11; +6.63	+0.23	2.08	+0.55
RMSE (m/s)	44 (13.8)	0.57–3.95	2.07	0.64	1.96
R	81 (13.8)	0.380–0.914	0.771	0.150	0.840
Inland					
Height (m)	305 (47.9)	10–200	71.8	56.1	50.0
MB (m/s)	179 (41.5)	−0.65; +2.628	+0.37	0.68	+0.27
RMSE (m/s)	110 (34.5)	0.57–2.44	1.57	0.39	1.53
R	289 (49.4)	0.635–0.940	0.843	0.062	0.855
Mountainous					
Height (m)	86 (13.5)	10–200	80.6	52.1	80.0
MB (m/s)	74 (17.2)	−4.36; +1.88	−0.57	1.33	−0.40
RMSE (m/s)	63 (19.7)	0.56–5.24	2.17	1.03	1.91
R	83 (14.2)	0.390–0.909	0.756	0.135	0.820
All					
Height (m)	637 (100)	10–200	59.4	55.5	42.0
MB (m/s)	431 (100)	−1.66; +2.00	+0.14	0.74	+0.08
RMSE (m/s)	319 (100)	1.00–4.68	1.88	0.62	1.76
R	585 (100)	0.450–0.945	0.827	0.095	0.853

^a Statistics are calculated at 95% confidence level after bootstrap resampling.

^b For each location type, values in brackets are normalized to the overall samples.

above ground. As expected, wind speed uncertainties (Fig. S4a) were mostly assessed based on 10-m monitors (40.7%), particularly located offshore (20.4%) again due to the availability of large buoy networks. Across the higher elevations (up to 300 m) inland monitors largely prevailed thanks to the availability of several met masts and LIDAR profilers. A remarkable 20.7% studies were found assessing wind speed uncertainties between 100 and 300 m. Performances of reanalysis products in estimating AEY (Fig. S4b) were mainly found across the 60–100 m range typical of modern WT hub heights (52.7%), markedly at offshore (20.7%), inland (17.8%) and mountainous (11.8%) locations. Across the 100–150 m range typical of modern offshore WT hub heights, most AEY applications were expectedly found offshore (11.8%).

Fig. 1 shows the overall distribution of reanalysis products that assess wind speed (Fig. 1a) and AEY (Fig. 1b), while Fig. S5 and Fig. S6 show the corresponding distribution detailed by location type. ERA5 (19.6%) and MERRA2 (18.1%) were the products most frequently applied for calculating wind speed (Fig. 1a), particularly over the most challenging coastal (Fig. S5b) and mountainous (Fig. S5d) locations, while application of earlier products (e.g. CFSR and R2) was quite frequent over offshore (Fig. S5a) and inland (Fig. S5c) locations. Overall, AEY uncertainties were assessed for only six reanalysis datasets (Fig. 1b), with ERA5 being the predominant choice (46.8%). The latter was the only product applied offshore (Fig. S6a) and the one mainly applied on mountainous locations (Fig. S6d). The European regional reanalysis products COSMO-REA2 and COSMO-REA6 were also frequently applied (23.1% on aggregate, Fig. 1b), particularly on coastal (Fig. S6b) and inland (Fig. S6c) locations.

4.2. Overall scores by location and height

4.2.1. Wind speed

The statistics of all reanalysis products assessed in calculating wind speed by location type are summarized in Table 4 through the MB, RMSE and R values. Fig. 2 shows the corresponding boxplots as a function of location and height. A bootstrap resampling procedure was applied to each dataset (given by each metric and location type), and then the corresponding confidence interval at 95% level was calculated. For each dataset, the latter is given by the corresponding range reported in Table 4, which also applies to the values reported in Table 6, and to the boxplots plotted in Figs. 2 and 4–7. Note that in figures no boxplot is created for zero size samples.

Overall, the reanalysis products show a slight median wind speed over-estimation (+0.08 m/s), which combines with a median RMSE of 1.76 m/s, and a time variation (which is generally an hourly variation, see Table 3) marked by a median R of 0.853 (Table 4). Assessed at 95% confidence level, intervals were found between −1.66 and +2.00 m/s for MB, 1.00–4.68 m/s for RMSE, and 0.450 to 0.945 for R. These results confirm that reanalyses are generally able to suitably reproduce wind speed variation over time, while they are not similarly able over space [10]. Wind speed may be significantly biased, also when applying a thorough spatial interpolation, whose importance has been found to increase with reanalysis spatial coarseness [45]. Thus, a proper calibration (or bias correction) should be addressed before using reanalysis data [40]. According to Ref. [10], the main reasons for this bias include systematic errors in the underlying forecasting model, spatial coarseness, and inadequate wind speed observations in the assimilation scheme.

Indeed, reanalysis bias correction has recently become a major research line. As shown, e.g., by Ref. [65], most studies have focused on MERRA [102–105] and MERRA2 [10,37,40,106] datasets, which generally return significantly biased wind speed estimates due to the coarse spatial resolution of their underlying models. To a lesser extent, bias correction also involved CFSR [107] and ERA5 [31,37,40,42]. Bias correction is generally performed using historical wind speed or AEY observations (e.g. Refs. [10,37,104]) or, more conveniently, estimations provided by the GWA (e.g. Refs. [40,42,44,106]). GWA v. 3.1 is a worldwide atlas that provides free mapping of onshore and offshore wind resources at 250-m grid spacing at heights of 10, 50, 100, 150 and 200 m [108]. Estimations by the GWA are derived from a 3-level model chain including a global forcing (~31 km) by the ERA5 reanalysis, a mesoscale downscaling (3 km) by the WRF model [21], and an ultimate microscale downscaling (250 m) by the WASP model [109]. An inconsistency often detected within such procedures is an uncritical application of reanalysis bias correction by geographical area such as country or region (e.g. Refs. [10,40,102,103]). Before applying any bias correction, a critical understanding of the universal features accounting for reanalyses performances would be preferable, for example by assessing uncertainties as a function of site characteristics and height above the ground.

The best reanalysis scores are achieved at offshore sites, as wind speed is slightly under-predicted (median MB = −0.10 m/s) and its time variation is well reproduced (median R = 0.891, Table 4). Corresponding confidence intervals range between −1.63 and +1.54 m/s for MB, and 0.572 to 0.956 for R. In terms of median values, wind speed is substantially unbiased up to 60 m, while at most it is under-predicted by −0.42 m/s across the 150–300 m range (Fig. 2a). Notably, among all locations, offshore applications return the lowest RMSE median values between 20 and 100 m (1.35 m/s, Fig. 2b), and the highest R median values between 10 and 150 m (up to 0.950, Fig. 2c). Concurring with several Authors (e.g. Ref. [74]), wind speed bias is lower over the sea than land. Particularly over large areas relatively far from the coast, offshore winds are more uniform and less turbulent, so they can be accurately reproduced using reanalysis data [93]. Maritime sites are quite homogeneous environments in terms of wind resource, thus

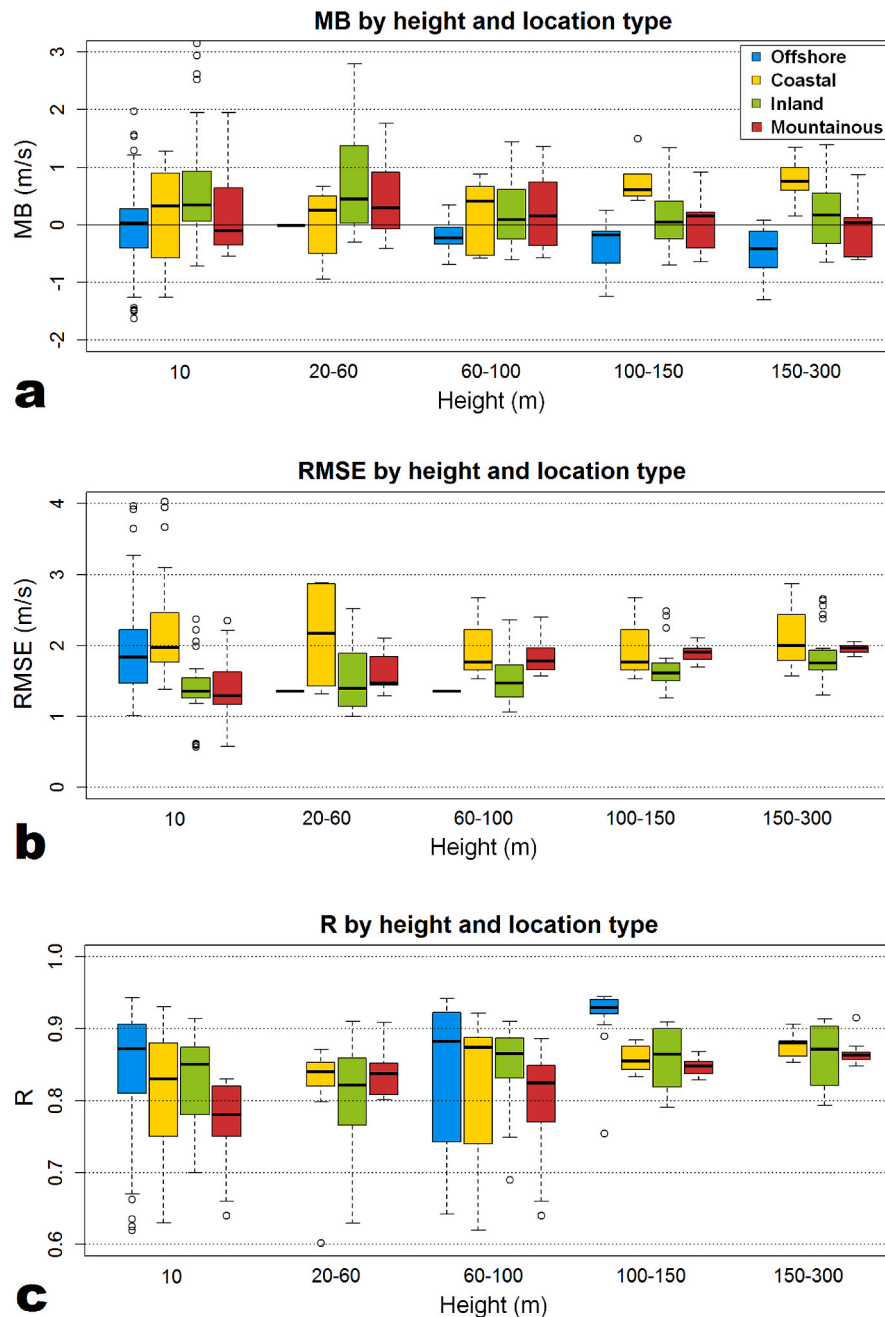


Fig. 2. Boxplots of (a) mean bias (m/s), (b) root mean square error (m/s), and (c) correlation coefficient in predicting wind speed per height and location type by all reanalysis products. Statistics are calculated at 95% confidence level after bootstrap resampling: the confidence intervals are given in the “Range” column of Table 4.

proving to be the “simplest” locations to be addressed, even by the coarser reanalysis products. It should be noted, however, that these offshore wind speed scores are predominantly obtained at an elevation of 10 m (Fig. S4a). Rather low uncertainties are also found at inland (mostly flat) locations, where a mere +0.27 m/s median wind speed over-prediction and the lowest overall RMSE median value (1.53 m/s) are returned. Median MB values are particularly low (at most +0.27 m/s) between 60 and 300 m, where median RMSE values are not higher than 1.74 m/s. Time variation is also well captured, with R showing a 0.855 median value with an overall range of 0.635–0.940. These statistics are particularly meaningful not only because inland sites largely prevail among all reanalysis applications (46.7%, Fig. S3a), but also as they are mostly referred to elevations between 20 and 100 m (Fig. S4a). Wind speed discrepancies are higher on both coastal and mountainous locations, affected by the highest median over-estimations (+0.55 m/s)

and under-estimations (−0.40 m/s), respectively, as well as higher RMSE and lower R median values. Fewer applications have been found on these more challenging sites (Fig. S3a): coastal winds are mostly assessed at 10 m, while mountainous winds are at between 60 and 100 m (Fig. S4a). Reanalysis products generally under-estimate wind resource on mountainous locations, also showing a relevant range (−4.36 to +1.88 m/s). At almost all heights, time variation is reproduced less accurately than elsewhere, as median R values range 0.760–0.860 (Fig. 2c). Because of their coarse spatial resolution, reanalyses are unable to resolve the complex terrain of mountainous regions [10]. Since providing winds averaged over a large area (grid-averaged winds), reanalyses tend to under-estimate speed-up effects occurring at sites such as hilltops or ridges, where winds are usually stronger than the surroundings [110]. Overall, the worst performances are found on coastal sites, particularly in terms of discrepancy of wind speed, as

Table 5
Statistics by location type of AEY scores achieved from all reanalysis products^a.

Locations	Valid data sample (%) ^b	Range	Mean	Std. dev.	Median
Offshore					
Height (m)	57 (35.2)	58–165	93.5	18.2	98.0
PE (%)	57 (35.2)	−25.9; +11.2	−4.2	6.9	−4.7
Coastal					
Height (m)	23 (14.2)	10–250	120.9	78.7	110.0
PE (%)	23 (14.2)	−20.8; +200.0	+44.4	56.2	+20.0
Inland					
Height (m)	63 (38.9)	10–200	86.5	56.3	80.0
PE (%)	63 (38.9)	−34.4; +268.9	+28.9	53.6	+10.0
Mountainous					
Height (m)	22 (13.6)	50–100	91.6	14.5	100.0
PE (%)	22 (13.6)	−93.6; +254.5	−4.5	85.7	−13.8
All					
Height (m)	162 (100)	10–250	93.9	48.5	98.0
PE (%)	162 (100)	−80.1; +200.0	+14.0	43.7	+2.8

^a Statistics are calculated at 95% confidence level after bootstrap resampling.

^b For each location type, values in brackets are normalized to the overall samples.

shown by the wide MB overall range (−11.11 to +6.63 m/s). On coastal locations, median RMSE is the highest (up to 2.17 m/s, Fig. 2b) between 10 and 60 m. These results confirm that coastal regions pose a modelling challenge compared to inland/flat or open sea locations. Complex flow regimes (including sea-land breezes) often develop there due to the coastline transition, characterized by roughness and topography discontinuities as well as thermal gradients (and hence strong atmospheric stability variations) [24]. Furthermore, as shown by Ref. [94], reanalysis products (particularly the coarser ones) are hampered by land masking issues that reduce their accuracy in simulating coastal winds.

4.2.2. Energy yield

The statistics of reanalysis performances in calculating AEY by location type in terms of PE values are summarized in Table 5, while PE boxplots as a function of location and height are plotted in Fig. 3. Again, the PE statistics by location type have been calculated at 95% confidence level after applying a bootstrap resampling. Their confidence interval is given by the range reported in Table 5, which also applies to the values reported in Table 7, and to the plots in Figs. 3 and 8.

Energy production is generally over-estimated by the reanalysis products, by +14% on average and +2.8% in terms of median values (Table 5). Again, applications were prevalent on inland (37.6%) and offshore (34.1%) locations (Fig. S3b). Scores on offshore sites are better than elsewhere, as AEY is on average under-predicted by −4.2% (Table 5). Remarkably, unlike the case of wind speed applications (see section 4.2.1), these AEY offshore scores are predominantly achieved at an elevation between 60 and 100 m (Fig. S4b). AEY uncertainties are greater at inland locations, with an overall mean over-prediction of +28.9%, mainly due to uncertainties across the 20–60 m (+44.2%) and 60–100 m (+34.3%) height ranges (Fig. 3 and S4b). Power output is generally under-estimated at mountainous sites (median PE = −13.8%), and over-estimated at coastal sites (median PE = +20%, Table 5). On mountainous locations, these AEY scores are mostly obtained at an elevation between 60 and 100 m, while on coastal locations they are fairly homogeneously distributed across all height ranges (Fig. S4b).

Table 6
Median values by reanalysis product and location type of wind speed scores achieved from all reviewed studies^{a,b,c}.

Locations	Reanalysis product															
	R1	R2	ERA40	ERA-Interim	CFSR	CFSv2	MERRA	MERRA2	ERA5	NARR	COSMO-REA6	COSMO-REA2	HARMONIEv1	UM	MERA	All
MB (m/s)																
Offshore	−0.35	+0.82	−0.36	+0.40	−0.03		−0.62	−0.44	+0.07	−0.28			−0.17			−0.10
Coastal				+0.47	+0.94		−1.16	+1.57	−0.50		+0.61	+0.50			−0.66	+0.55
Inland				+0.34			+2.58	+0.97	−0.09		−0.10	+0.33			+1.00	+0.27
Mountainous				+0.01			+0.84	+0.35	−0.88		−0.57	−0.50			−0.40	
All	−0.35	+0.93	−0.36	+0.33	+0.03		−0.59	+0.92	−0.08	−0.28	−0.10	+0.30	−0.17		−0.01	+0.08
RMSE (m/s)																
Offshore	2.05	3.92	5.10	2.46	1.83		2.08		1.45							1.86
Coastal				1.28	1.95			2.67	1.78		1.77				1.65	1.96
Inland				0.61				1.98	1.34		1.49	1.37			1.67	1.53
Mountainous				0.57				1.90	1.87		1.95	2.00				1.91
All	2.05	3.27	1.18	2.42	1.85		2.08	2.11	1.57		1.62	1.43			1.67	1.76
R																
Offshore	0.840	0.768	0.885	0.894	0.894	0.670	0.897	0.844	0.870	0.646	0.943		0.941	0.945		0.891
Coastal				0.818	0.505		0.829	0.827	0.750		0.873	0.840			0.890	0.840
Inland		0.761		0.870	0.841		0.841	0.850	0.860		0.875	0.846	0.879	0.895	0.860	0.855
Mountainous				0.765			0.590	0.808	0.837		0.832	0.775				0.820
All	0.840	0.762	0.885	0.870	0.866	0.663	0.841	0.843	0.858	0.646	0.870	0.820	0.905	0.906	0.870	0.853

^a Median values are calculated at 95% confidence level after bootstrap resampling; the confidence intervals are given in the “Range” column of Table 4.

^b For each reanalysis and location, scores finer than overall scores (“All” column) are highlighted in bold.

^c Blank cells mean data not available.

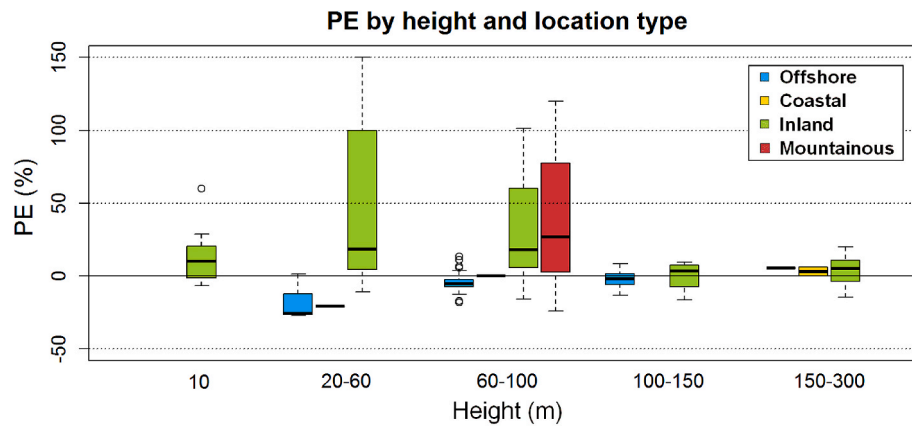


Fig. 3. Boxplots of percent error (%) in predicting AEY per height and location type by all reanalysis products. Statistics are calculated at 95% confidence level after bootstrap resampling; the confidence intervals are given in the “Range” column of [Table 5](#).

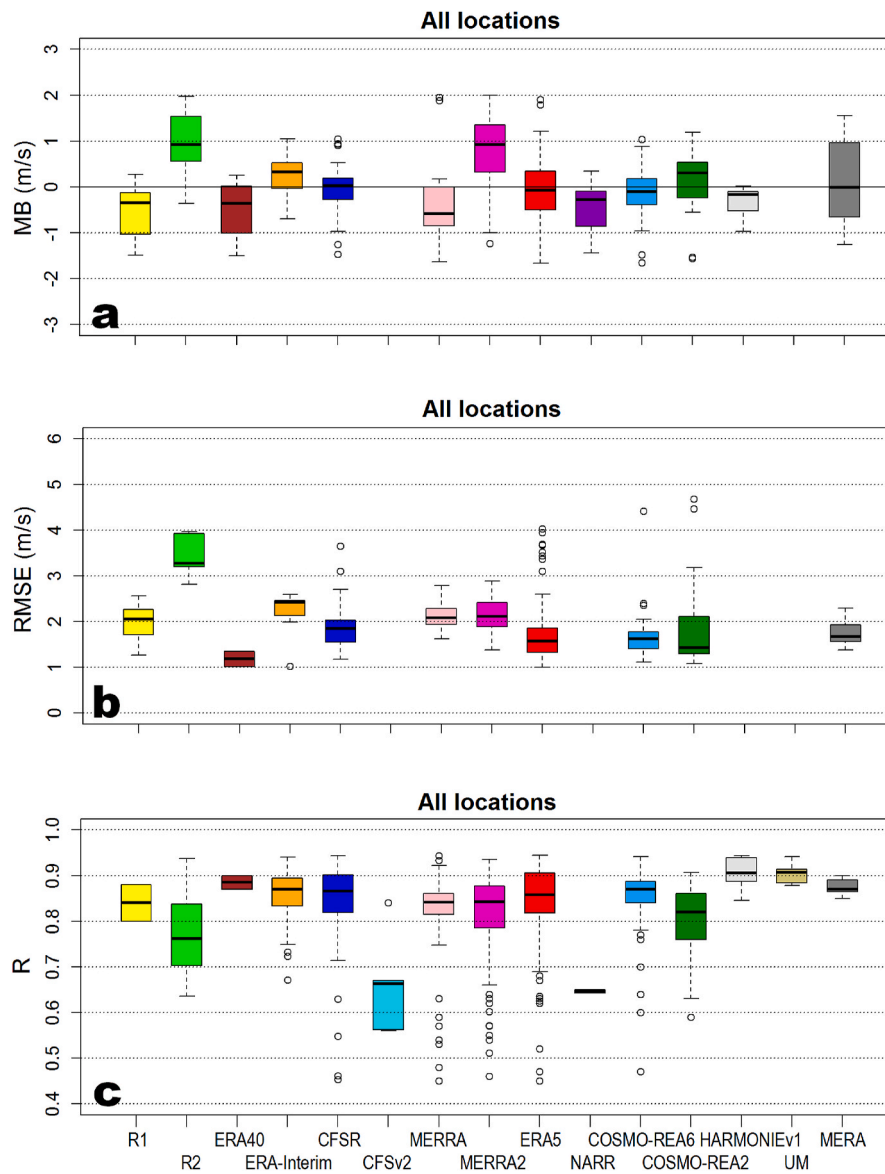


Fig. 4. Boxplots of (a) mean bias (m/s), (b) root mean square error (m/s), and (c) correlation coefficient in predicting wind speed by reanalysis product at all locations. Statistics are calculated at 95% confidence level after bootstrap resampling; the confidence intervals are given in the “Range” column of [Table 4](#).

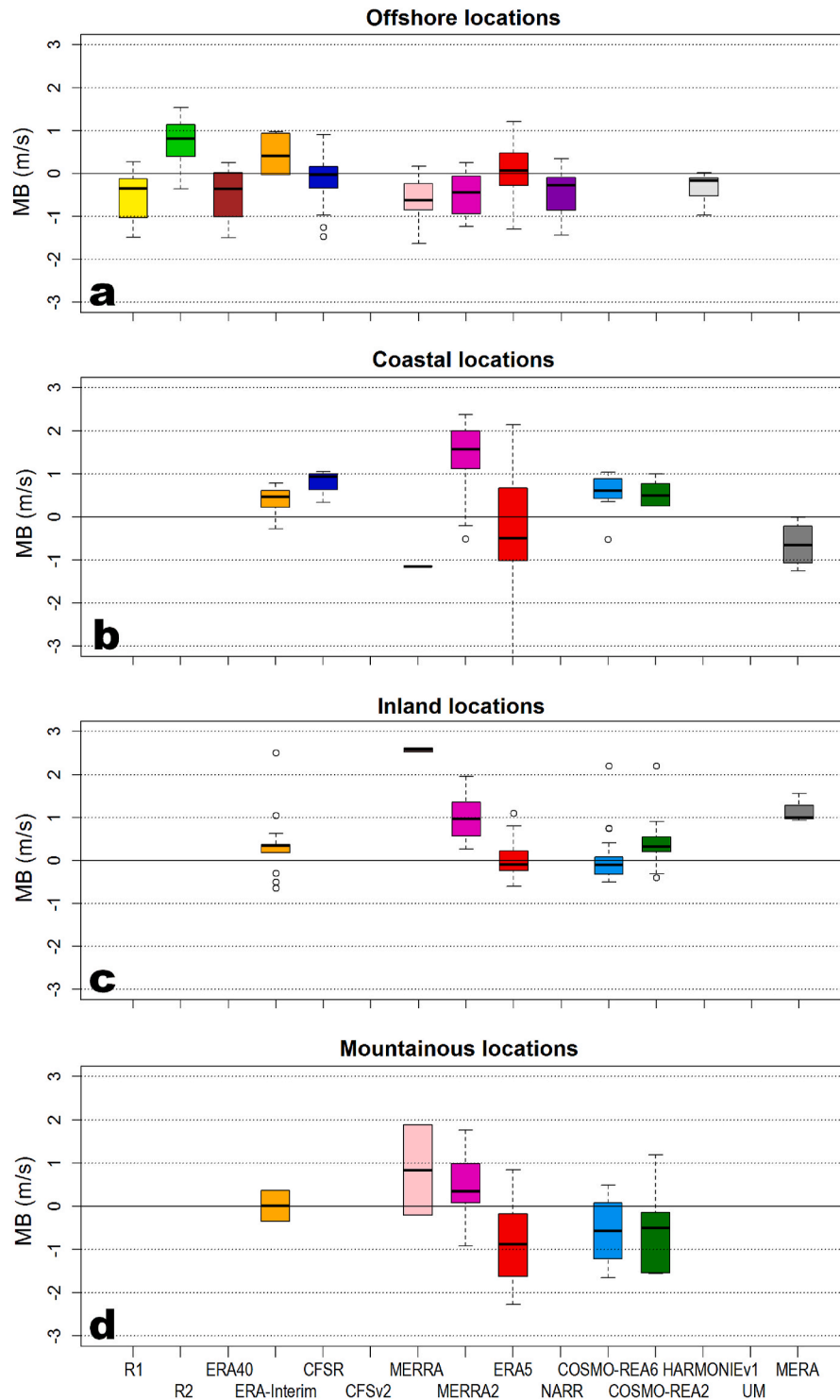


Fig. 5. Boxplots of mean bias (m/s) in predicting wind speed by reanalysis product at: (a) offshore, (b) coastal, (c) inland, and (d) mountainous locations. Statistics are calculated at 95% confidence level after bootstrap resampling: the confidence intervals are given in the “Range” column of [Table 4](#).

4.3. Detailed scores by reanalysis product

4.3.1. Wind speed

Wind speed uncertainties overall exhibited by each reanalysis product are plotted in [Fig. 4](#) in terms of frequency distribution of MB, RMSE and R. For each reanalysis, the boxplots of MB, RMSE and R scores as a function of location type are presented in [Figs. 5–7](#), respectively. [Table 6](#) summarizes the MB, RMSE and R median values by all reanalysis products assessed in calculating wind speed by location type. It is worth

noting that a time step coarser than 1-h was used for NARR (3-h) and for R2, ERA40 and UM (6-h), while this was often the case for ERA-Interim (3-h or 6-h) as well as R1 and HARMONIEv1 (6-h, [Table 3](#)).

Analysing [Fig. 4](#) and the “All” rows of [Table 6](#), CFSR and ERA5 are among the best-performing global reanalysis datasets, as they exhibit MB distributions tightly peaked around a near zero median value (Q2), rather low RMSE distributions with a still low interquartile range (IQR), and moderately high R distributions with Q1 greater than 0.818. ERA5 scores, however, are more meaningful due to their larger sample size

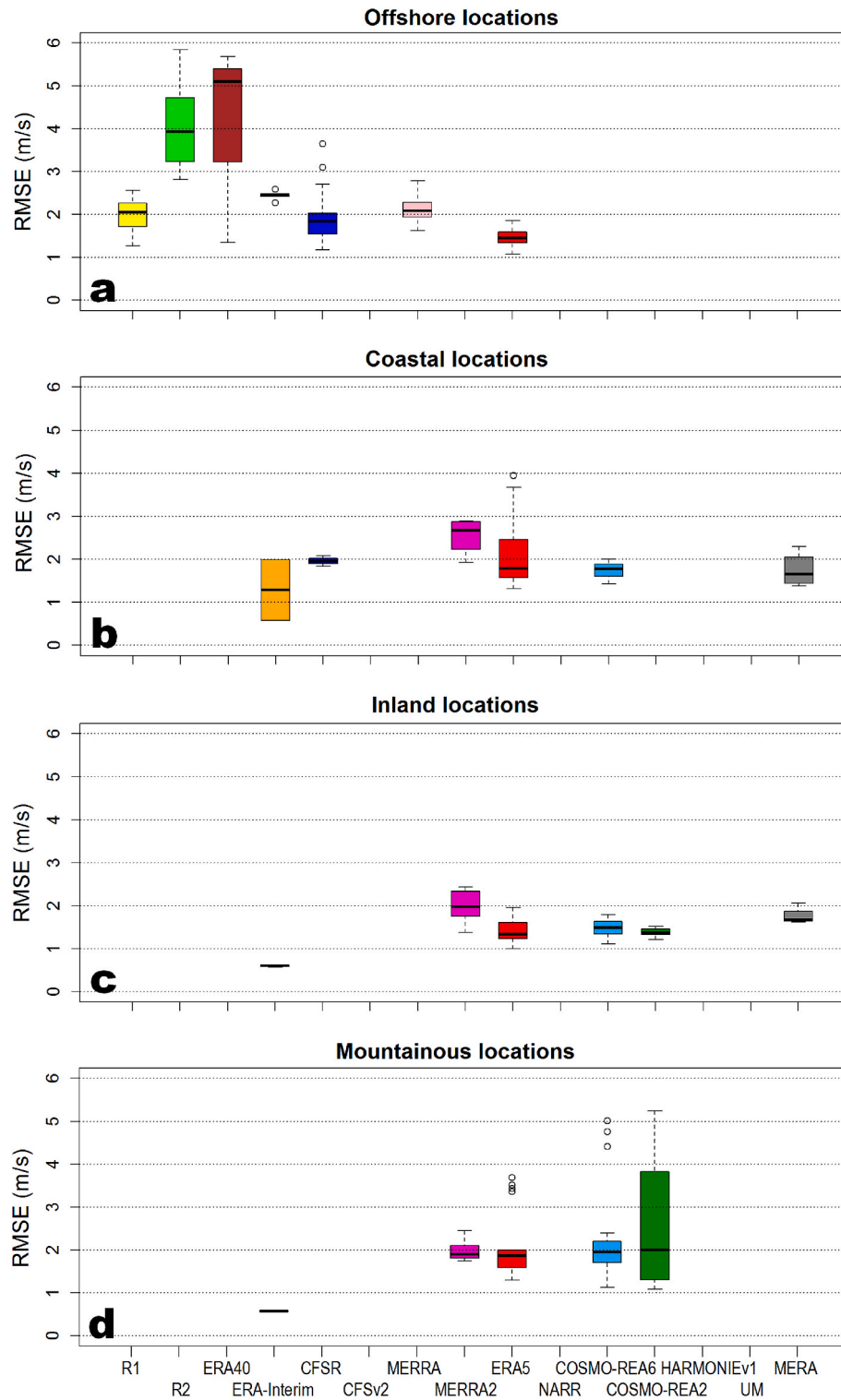


Fig. 6. Same as Fig. 5, but for root mean square error (m/s).

than CFSR (19.6 vs. 11.8%, Fig. 1a). Among the most frequently applied global reanalysis products, uncertainties are higher for MERRA and MERRA2: while their RMSE and R distributions are quite similar, MERRA returns a general under-prediction (IQR of MB between -0.06 and -0.84 m/s), while MERRA2 an over-prediction (IQR between $+0.34$ and $+1.35$ m/s, Fig. 4a). Wind resource is generally better predicted by the European regional reanalyses. COSMO-REA6 and COSMO-REA2, applied on aggregate in 19.5% of studies (Fig. 1a), return narrow MB distributions very close to zero (at most with median MB of $+0.30$ m/s,

Table 6), rather low RMSE distributions, and rather high R distributions. Consistently with [84], although they have different resolution, no significant difference in accuracy has been identified between COSMO-REA2 and COSMO-REA6, albeit the latter slightly outperforms the former, particularly in capturing wind speed time variation (Fig. 4c). This outcome suggests that – at least when dealing with high-resolution regional reanalysis products applied on quite simple locations – a higher spatial resolution does not always lead to a better simulation. Overall performances are also remarkable for HARMONIEv1 and UM: although

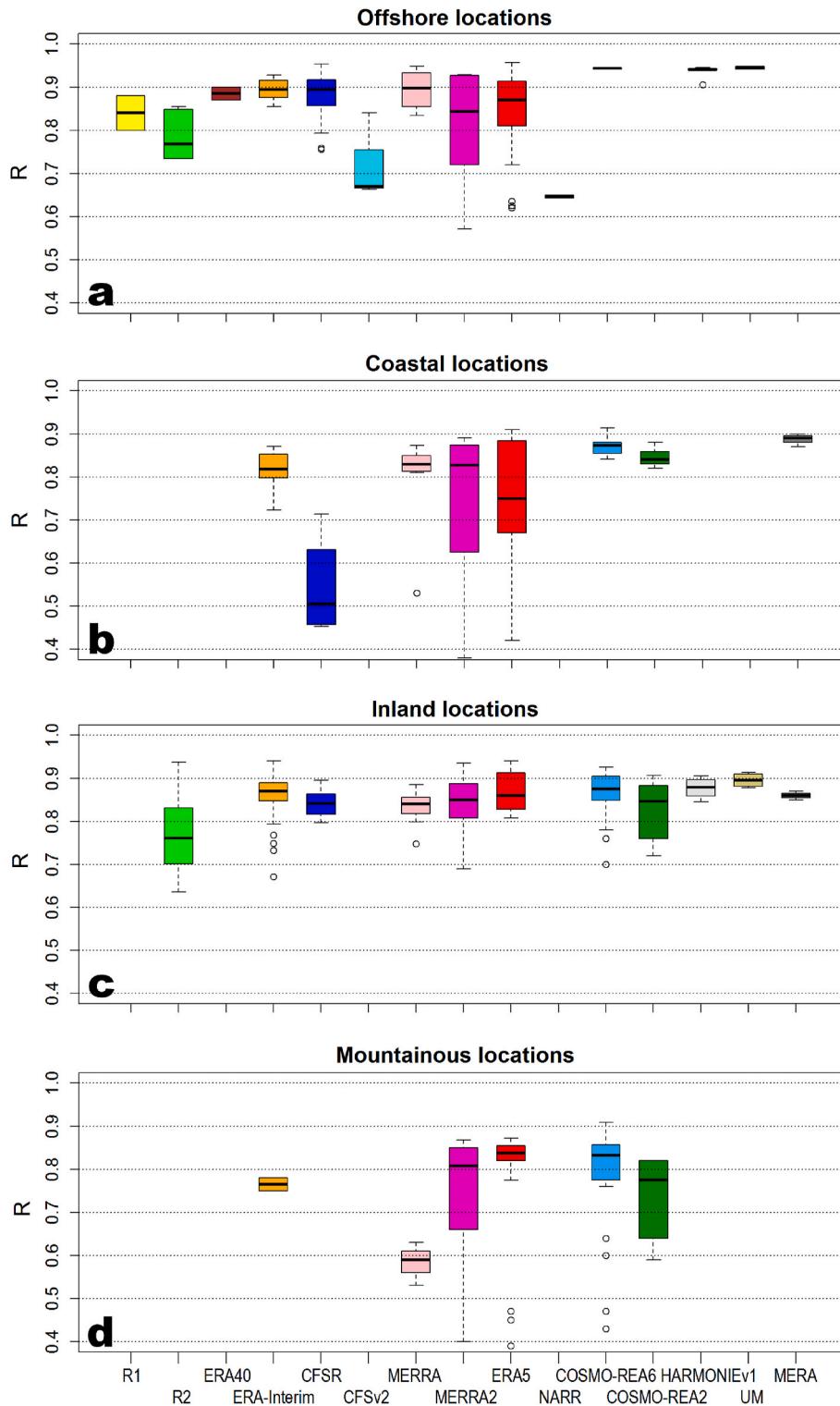


Fig. 7. Same as Fig. 5, but for correlation coefficient.

quite rarely applied (3%), they exhibit the highest R distributions (Q1 above 0.885).

The analysis by location type sheds more light on reanalysis uncertainties.

At offshore sites, wind speed is accurately predicted by most reanalyses, with only the IQRs of R1 and R2 (slightly) falling outside the ± 1 m/s MB range (Fig. 5a), and only (again) R2 and ERA40 exhibiting very large and high RMSE distributions (Q3 up to 5.39 m/s, Fig. 6a).

Confirming findings from Ref. [12], the scores of R2 are worse than those of its predecessor R1 (Table 6). Overall, CFSR and ERA5 return the best offshore performances, as they show both the lowest discrepancies (Fig. 5a) and lowest RMSE values (Fig. 6a), as well as quite high R values (Fig. 7a). Notably, wind resource time variation is better predicted by CFSR than its successor CFSv2. The scores for NARR are worse, particularly for R (Fig. 7a): as suggested by Ref. [45], wind speed uncertainties by NARR on offshore sites may be due to the underlying model and

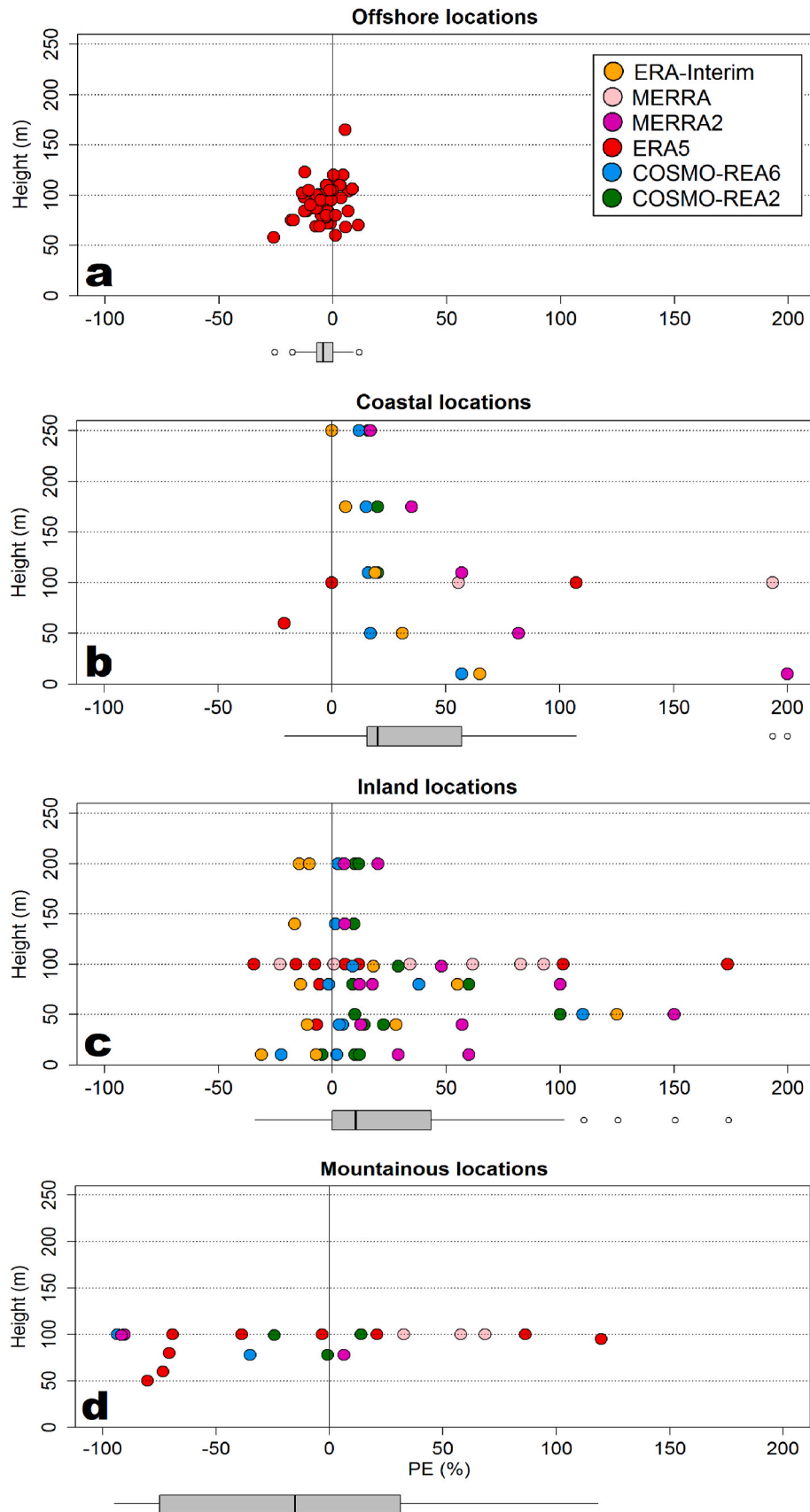


Fig. 8. Percent error (%) in predicting Aey by the reanalysis products at: (a) offshore, (b) coastal, (c) inland, and (d) mountainous locations. The overall boxplots by location are also shown. Statistics are calculated at 95% confidence level after bootstrap resampling: the confidence intervals are given in the “Range” column of Table 5.

−0.18 m/s, Fig. 5d). ERA5 scores are achieved at a median height of 87.5 m, thus not exactly the 100-m native height of its wind data. However, with an MB variation range ($\mu \pm \sigma$) of -0.86 ± 0.84 m/s, ERA5 scores are again better than those reported by Ref. [27] after comparison vs. 85 masts located on highly complex sites in Europe ($\mu \pm \sigma = -2.64 \pm 1.17$ m/s).

On coastal locations, disagreeing with [111], ERA-Interim is the best performing reanalysis in predicting wind speed magnitude (Figs. 5b and 6b), outperforming its successor ERA5 also in predicting wind speed time variation (Fig. 7b). ERA-Interim, however, was applied over significantly fewer case studies than ERA5 (Fig. S5b). Unlike elsewhere, the use of regional datasets implementing a proper downscaling of global reanalysis wind speed time series is recommended on coastal sites. There, the most frequently used ERA5 (30.7%) and MERRA2 (27.3%) datasets are outperformed by COSMO-REA6 and COSMO-REA2, whose distributions exhibit small IQRs for all three metrics with particularly good median values. Scores are markedly poor for MERRA2, which, due to its low spatial resolution, fails to resolve the sea-land differences on coastal locations [71]. Here, scores by MERRA are slightly better than those by MERRA2. Conversely, due to their higher resolution, regional datasets are able to mitigate the land masking effects across these areas [94]. It is noteworthy that these coastal scores are obtained at median heights of 10 m (ERA5 and MERRA), 50 m (ERA-Interim), 80 m (MERRA2), 105 m (COSMO-REA6) and 110 m (COSMO-REA2). Thus, scores by COSMO-REA6 and COSMO-REA2 (which are quite similar) are even more meaningful as they are achieved at more challenging elevations.

Overall, it is confirmed that ERA5 outperforms MERRA2 in reproducing both wind speed magnitude and its time variation on offshore [45,80], inland [37,71] and coastal [71] locations, while the opposite applies on mountainous locations [13] (Table 6). More in general, current scores reflect all the improvements implemented in reanalyses across the years, such as larger number and wider spectrum of assimilated observations or improved forecasting model [113]. In particular, various Authors (e.g. Refs. [71,112]) highlighted the data assimilation scheme being pivotal, suggesting that reanalyses implementing the newer 4D-Var scheme have a major advantage over those implementing the older 3D-Var scheme (Table 1). It should be borne in mind that most regional reanalyses (COSMO-REA2, COSMO-REA6 and UM) assimilate surface winds from land stations [2,83], while no global reanalysis does [9,12] (see section 2). According to Ref. [10], this lack is a primary source of error by global reanalyses. As pointed out by Ref. [12], this is an issue particularly over areas with inhomogeneous terrain (mountainous or coastal regions), where winds undergo strong local influences that global reanalyses fail to properly capture. However, clear reasons lie behind the choice of not assimilating wind observations from surface land stations by global reanalyses (a choice also applying to NWPs). On the one hand, it was reported that 10-m wind observations over land have little impact on wind forecast; on the other, assimilation of 10-m land wind observations generally causes problems to the forecasting model in consistently adapting the boundary layer, resulting in too strong wind forecast at night [114].

Another key goal fostering application of the reanalysis products is to provide accurate initial and boundary conditions to MWP models for wind simulation. In this regard, for example, after testing datasets as R2, ERA-Interim, CFSR and MERRA, ERA-Interim was found as the most accurate product to force the WRF model, both onshore [113] and offshore [24]. A similar outcome was achieved by Ref. [115] after comparing R2 and ERA-Interim in forcing WRF over several coastal locations in the Persian Gulf. The Authors suggested that a strong reason supporting this outcome is the 4D-Var data assimilation system implemented in ERA-Interim with respect to 3D-Var implemented by the other products (Table 1). On the other hand, when reanalyses implementing the same 4D-Var scheme were compared, as was the case of ERA-Interim vs. its successor ERA5 over complex terrain in Norway, wind simulations by WRF forced by ERA5 proved to be finer [116]. This means that all

other advances implemented over the years played a role, which led, e.g., to choose ERA5 to force WRF to create (via WAsP) the New European Wind Atlas [27].

4.3.2. Energy yield

Fig. 8 presents the PE values in predicting AEY by height returned by the six reanalysis datasets as a function of location type. The same scores, aggregated for all location types, are presented in Fig. S7. Energy yield PE mean values by location type for each reanalysis product are summarized in Table 7.

Overall, the best scores are achieved by ERA5 and COSMO-REA6, as they exhibit mean PE values of −1.5% and +13.7%, respectively (Table 7). Scores from ERA5 are however more meaningful as it is by far the most frequently applied dataset (46.8%, Fig. 1b). AEY uncertainties are higher for COSMO-REA2 (mean PE = +17.8%), which is once again outperformed by COSMO-REA6. The accuracy of MERRA and MERRA2 is also lower in predicting AEY, as mean PE values of +59.8% and +50.8% are returned, respectively. Confirming outcomes from various Authors (e.g. Refs. [37,40]), ERA5 outperforms MERRA2 in predicting AEY over all location types.

As the only product applied offshore (Fig. S6a), ERA5 exhibits fine scores, as a slight mean AEY over-estimation is returned (PE = −4.2%, Table 7) with a small standard deviation (6.9%, Table 5). These scores mostly derive from elevations between 60 and 100 m (Fig. 8a and S4b) and are mostly achieved at a median height of 97 m, thus basically matching the 100-m ERA5 native wind data elevation. With AEY on average under-predicted by −3.7%, COSMO-REA2 is the best reanalysis dataset applied on mountainous locations (Table 7), where application heights are limited between 50 and 100 m (Fig. 8d). This outcome confirms findings from Ref. [84] that COSMO-REA2 clearly outperforms COSMO-REA6 only on highly uneven sites, so that a higher spatial resolution is more urgent over complex terrain locations. On the contrary, consistently with the literature (e.g. Ref. [85]), ERA5 exhibits a significantly wide (both positive and negative) discrepancy range ($\mu \pm \sigma = -12.0 \pm 74.2\%$), which is not dissimilar to that ($-69.4 \pm 24.9\%$) reported by Ref. [27] on highly complex sites in Europe. Notably, AEY is greatly under-predicted by MERRA2 (PE = −58.6%), while even more greatly over-predicted by MERRA (PE = +103.5%). As for wind speed prediction, ERA-Interim is also the best-performing dataset for energy production at inland sites: it shows a PE mean value (+11.1%) lower than those of both COSMO-REA6 (+13.8%) and COSMO-REA2 (+22.5%), and particularly of its successor ERA5 (+24.6%). Here, COSMO-REA6 is better than COSMO-REA2. Current ERA5 scores in predicting AEY on inland locations ($\mu \pm \sigma = +24.6 \pm 67.7\%$) are worse than those ($-20.5 \pm 20.5\%$) found by Ref. [27] over simple terrain sites in Europe after comparison vs. 110 masts at heights between 40 and 150 m. Overall, the highest reanalysis uncertainties are observed between 50 and 100 m, while they are lower further up (Fig. 8c). Here, MERRA2 improves the performances of its predecessor. AEY is largely over-predicted at coastal sites (Fig. 8b), where the lowest discrepancies are shown by COSMO-REA2 (mean PE = +18.7%) and COSMO-REA6 (+23.4%, Table 7). Here, COSMO-REA2 is slightly better than COSMO-REA6. As on inland locations, ERA-Interim ($\mu \pm \sigma = +24.2 \pm 25.8\%$) is also slightly more accurate than ERA5 (+28.8 ± 68.7%) and significantly so than MERRA2 (+78.2 ± 72.3%). Uncertainties from the latter, however, are lower than those from its predecessor. Again, on coastal sites reanalysis uncertainties decrease as application height increases, particularly above 150 m (Fig. 3).

In general, if comparing reanalysis scores in estimating wind speed and AEY (as is the case of ERA5 in Figs. 5a and 8a), an over-estimation of the former may be contrasted by an under-estimation of the latter, or vice versa. This can happen for various reasons, including differences in data sample or in reference height (generally higher for AEY estimation), but mostly because only a selected range of wind speed values (those comprised between cut-in and cut-off wind speeds of WT) is used for AEY estimation, while the full range is used for wind speed

estimation.

Although not directly providing air density, reanalyses generally deliver time-varying values of air temperature, pressure, and humidity, which can be combined altogether to calculate time-varying air density [117]. For wind energy purposes, this represents a clear benefit than considering a constant (yearly) air density value. In recent years, reanalysis datasets have been used to assess temporal variations of air density, both at local [117–120] and global [121] scale, and thus to quantify the resulting deviations in wind power density and AEY. For example, over the Black Sea region [118], air density changes relative to the annual average were found ranging between -4% in summer to $+4\%$ in winter. Offshore in Lebanon [119], seasonal oscillations in power density ranging from -4% in summer to $+3\%$ in winter were observed. Offshore in northeastern Scotland [120], summer-winter power density variations up to 2–3%, corresponding to AEY variations up to 1%, were reported. The Authors also highlighted that short-term air density variations distort the real WT power curve in its U^3 zone, with effects similar to pitch misalignments or control and regulation faults. More in general, within their study worldwide [121], the Authors observed the highest seasonal air density changes occurring over northern extra-tropical continental regions (North America, Europe and Asia), with increases up to $+10\%$ during cold seasons and negative changes up to -10% during warm seasons. Over these areas, they reported AEY variations from -6 to $+6\%$ from summers to winters. Conversely, a smaller impact of air density changes was observed over tropical areas. Since water vapour is less dense than dry air, the density of moist air is lower, and so is power density [117]. Although the influence of humidity on air density is generally recognised as small and in most cases negligible [120], Floors and Nielsen [117] highlighted that air density (and thus AEY) may be slightly over-predicted when the effects of humidity are ignored. It should be noted that the majority of works calculating AEY reviewed in the present study ([28,70,77,81,85], Table 3) assumed (either directly or indirectly) a constant value for air density, which thus represents a further source of error in AEY prediction. Only in Ref. [14] the hourly air density values were considered, being calculated based on air temperature and pressure 1-h values retrieved from ERA5, while humidity was ignored.

4.4. Comparison of reanalysis scores against newer studies

Current reanalysis scores have been checked through a comparison against those returned by a number of articles published after the query running date (Nov. 16, 2021, see section 4.1) that report applications satisfying the criteria set in Table 2. The comparison has been performed by product and location type, so that it was only applied when both these factors matched. Only reanalysis applications for wind speed prediction have been found. The summary of the scores derived from these studies is summarized in Table S1.

Kanwal et al. [122] assessed 10-m wind speed by R1, R2 and CFSR using 2–3 years of 6-h observations collected by met masts deployed on three coastal and three inland locations in Pakistan. On coastal sites, current median values for CFSR (Table 6) are slightly worse than those reported by Ref. [122] (Table S1). On inland sites, present IQRs of R return comparable scores for R2 (0.70–0.83), whereas better scores for CFSR (0.82–0.86, Fig. 7c). Furthermore, on coastal sites Ref. [122] achieved R2 scores worse than those by R1, while the opposite applies on inland sites (Table S1).

De Assis Tavares et al. [123] performed an intercomparison of CFSv2, MERRA2 and ERA5 against 1 year of 1-h wind speed observations collected at 10 m on one offshore and six coastal sites located in the state of Rio de Janeiro (Brazil, Table S1). Current offshore scores are better for all products in terms of median R (0.670–0.870) and median MB for ERA5 ($+0.07$ m/s), while they are worse for median MB for MERRA2 (-0.44 m/s) and median RMSE for ERA5 (1.45 m/s). Present scores on coastal sites are slightly better for ERA5, whereas slightly worse for MERRA2 (Table 6).

Potisomporn and Vogel [124] conducted a validation of MERRA2 against met mast observations over four offshore wind farms in the UK using 18 years (2000–2017) of 1-h wind speed data at 100 m. Current offshore scores by MERRA2 are slightly worse both in terms of MB (Fig. 5a) and R (Fig. 7a).

Ahmad et al. [125] assessed 100-m wind speed by ERA5 on a coastal site in Pakistan using 2 years (May 2009 to Apr. 2011) of 1-h observations collected from a met mast. Their scores are better than MB values currently achieved for ERA5 (Fig. 5b), while they are comparable in terms of RMSE (Fig. 6b).

Pronk et al. [126] compared one year of 1-h wind profiles from ERA5 to those observed in the US by two LIDARs deployed offshore and one installed on an inland site. Offshore, two floating LIDARs mounted on buoys off the coast of New Jersey were considered, providing observations from Sep. 2019 to Aug. 2020 at heights between 60 and 200 m. At the flat inland site of SGP (Oklahoma), LIDAR data from 1 Jan. to Dec. 31, 2018 at six heights between 65 and 195 m were used. Both on offshore and inland locations, present ERA5 scores are better in terms of MB, while they are slightly worse for R (Table 6).

An example of regional reanalysis validation is presented by Hadjipetrou et al. [127], who compared 10 years (Jan. 2009 to July 2019) of 6-h wind speed by HARMONIEv1 against five met stations located on coastal sites in Cyprus, ultimately reporting remarkably biased predictions (Table S1). Unfortunately, no HARMONIEv1 application on coastal sites are available for comparison here (Fig. S5b).

5. Conclusions

In the last few years, the use of global reanalysis datasets for direct wind resource assessment has received a significant boost, particularly thanks to availability of wind records at elevations suitable for wind energy purposes, such as in the case of MERRA2 (50 m) and ERA5 (100 m). Since its release in 2019, for example, ERA5 has been used in countless applications as a direct input for wind resource assessment (e.g. Refs. [4,29,31–33,37]). Several regional reanalysis products, dynamically downscaling global reanalysis data to specific areas, have also been developed to gain higher spatial resolution, and thus better wind resource assessment. This paper presents a critical review on the uncertainties assessed in the literature by reanalysis datasets in estimating wind resource and energy yield. A total of 40 articles were reviewed, assessing scores achieved by 15 (9 global and 6 regional) reanalysis products on 322 different locations over 30 countries worldwide.

Confirming findings from the literature (e.g. Ref. [10]), reanalyses are generally well capable of predicting the wind resource pattern over time, while they are not similarly able over space. As well-recognised (e.g. Ref. [93]), reanalysis data (particularly by ERA5) are sufficiently reliable to assess both wind resource and AEY on offshore locations. Rather low uncertainties were also detected at inland (basically flat) locations, where most wind speed applications were found (46.7%). Here, ERA5 is again the best dataset in assessing wind resource, while in predicting AEY it is outperformed by COSMO-REA6 and ERA-Interim. Consistently with various Authors (e.g. Refs. [10,110]), reanalysis products generally under-estimate both wind resource and AEY on mountainous regions. Because of their coarse spatial resolution, they fail to resolve the complex topography of these locations as grid-averaged winds tend to under-estimate speed-up effects occurring there. MERRA2 returns the lowest wind speed discrepancies on mountain locations, while AEY is best predicted by COSMO-REA2. The worst overall performances were found on coastal sites. On these regions, strong roughness and topographical discontinuities as well as thermal gradients [24], associated with land masking issues [94], prevent reanalysis products from accurately simulating wind resource. Here, both wind speed and AEY are over-estimated. Scores achieved by ERA-Interim, the best among global products, are improved by regional datasets as COSMO-REA6 and COSMO-REA2. In general, reanalysis performances

achieved in the present review showed patterns basically in line with those reported by several more recently published articles.

Comparison between older and newer versions of reanalysis datasets (e.g., MERRA vs. MERRA2, ERA-Interim vs. ERA5, COSMO-REA6 vs. COSMO-REA2) allowed to address to what extent all technical advances implemented over the years (finer spatial and temporal resolution, larger number and wider spectrum of assimilated observations, improved data assimilation system, improved forecasting model, wind speed availability at higher elevation, etc.) actually result in higher prediction accuracy. Surprisingly, scores by the newer versions were not always better than those by the older ones. In many cases, however, the improvement from the newer products was found – rather in accuracy itself – in pushing a given accuracy towards higher (more challenging) elevations, thus more suitably meeting the needs of the modern wind power industry.

To summarise, on offshore and flat onshore locations the use of global reanalysis data (particularly by ERA5) should be preferred over higher-resolution regional products – at least when assessing wind resource potential of a country/region for strategic planning [34] or future projected scenarios [128] – as they are more efficient in terms of time, cost, computation resources and data requirements. Conversely, regional reanalyses (particularly COSMO-REA6) should be used – clearly, where coverage is available – to address fine-scale aspects involving WTs or wind farms such as, e.g., vertical wind gradients or wind speed changes per time interval (ramp rates) [81]. The use of regional reanalyses is also recommended over the more challenging coastal and mountainous sites, where downscaling of global reanalyses is needed to adequately resolve detailed terrain features as well as land use discontinuities and strong stability variations [94]. Thus, a bias correction of reanalysis data, e.g. addressed by using high-resolution outputs from the GWA, may (possibly) be implemented on offshore or flat onshore sites, while it is strongly recommended when dealing with coastal or mountainous sites. It must be mentioned, however, that efforts to improve global reanalysis products should be made at the beginning (rather than at the end) of the entire flowchart. For example, while most regional reanalyses assimilate surface winds from land stations [2,83], no global reanalysis does [9,12], resulting in a primary source of error [10]. In view of future reanalysis improvements, it is desirable that advances in forecasting models will enable current problems in assimilating surface land wind observations by the global reanalyses to be overcome. However at ECMWF, for example, it is not certain that this will happen within the new ERA6 global reanalysis product [55]. Importantly, refinements in global datasets could also have a positive effect on regional products, whose accuracy strictly depends on the quality of the global reanalysis they use as boundary conditions [71]. Although reanalyses (indirectly) deliver time-varying air density values, this review showed that in AEY prediction only one out of the selected studies took advantage of this opportunity rather than considering a constant air density value. As demonstrated by several Authors (e.g. Ref. [121]), conversely, this results in a further source of error in AEY prediction.

Credit author statement

Giovanni Gualtieri performed the whole work.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data from reanalysis products reviewed in the present paper may be downloaded at the following addresses (all accessed December 16, 2021):

- R1: <https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html>.
 R2: <https://psl.noaa.gov/data/gridded/data.ncep.reanalysis2.html>.
 CFSR: <https://climatedataguide.ucar.edu/climate-data/climate-forecast-system-reanalysis-cfsr>; CFSv2: <https://rda.ucar.edu/datasets/ds094.1>.
 MERRA: <https://climatedataguide.ucar.edu/climate-data/nasa-merra>.
 MERRA2: <https://climatedataguide.ucar.edu/climate-data/nasa-merra2-reanalysis>.
 ERA40: <https://apps.ecmwf.int/datasets/data/era40-daily/levtype=sfc>.
 ERA-Interim: <https://apps.ecmwf.int/datasets/data/interim-full-daily/levtype=sfc>.
 ERA5: <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=overview>.
 NARR: <https://climatedataguide.ucar.edu/climate-data/ncep-narr>.
 COSMO-REA6: <https://reanalysis.meteo.uni-bonn.de/?COSMO-REA6>.
 COSMO-REA2: <https://reanalysis.meteo.uni-bonn.de/?COSMO-REA2>.
 HARMONIEv2: <https://apps.ecmwf.int/datasets/data/uerra-harmonie-v1/levtype=sfc/stream=oper/type=an>.
 UM: <https://apps.ecmwf.int/datasets/data/uerra-um-4dvar/levtype=sfc/stream=oper/type=an>.
 MERA: <https://www.met.ie/climate/available-data/mera>.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rser.2022.112741>.

References

- [1] Samal RK. Assessment of wind energy potential using reanalysis data: a comparison with mast measurements. *J Clean Prod* 2021;127933.
- [2] Niermann D, Borsche M, Kaiser-Weiss AK, Kaspar F. Evaluating renewable-energy-relevant parameters of COSMO-REA6 by comparison with satellite data, station observations and other reanalyses. *Meteorol Z* 2019;28(4):347–60.
- [3] Pryor SC, Letson FW, Barthelmie RJ. Variability in wind energy generation across the contiguous United States. *J Appl Meteorol Climatol* 2020;59(12):2021–39.
- [4] Soares PM, Lima DC, Nogueira M. Global offshore wind energy resources using the new ERA-5 reanalysis. *Environ Res Lett* 2020;15:1040a2.
- [5] Gualtieri G. Improving investigation of wind turbine optimal site matching through the self-organizing maps. *Energy Convers Manag* 2017;143:295–311.
- [6] Gualtieri G. Surface turbulence intensity as a predictor of extrapolated wind resource to the turbine hub height: method's test at a mountain site. *Renew Energy* 2018;120:457–67.

- [7] Ramon J, Lledó L, Pérez-Zanón N, Soret A, Doblas-Reyes FJ. The Tall Tower Dataset A unique initiative to boost wind energy research. *Earth Syst Sci Data* 2020;12:429–39.
- [8] Jiménez PA, González-Rouco JF, Navarro J, Montávez JP, García-Bustamante E. Quality assurance of surface wind observations from automated weather stations. *J Atmos Ocean Technol* 2010;27(7):1101–22.
- [9] Fujiwara M, Wright JS, Manney GL, Gray LJ, Anstey J, Birner T, et al. Introduction to the SPARC reanalysis intercomparison project (S-RIP) and overview of the reanalysis systems. *Atmos Chem Phys* 2017;17(2):1417–52.
- [10] Staffell I, Pfenninger S. Using bias-corrected reanalysis to simulate current and future wind power output. *Energy* 2016;114:1224–39.
- [11] McKenna R, Pfenninger S, Heinrichs H, Schmidt J, Staffell I, Bauer C, et al. High-resolution large-scale onshore wind energy assessments: a review of potential definitions, methodologies and future research needs. *Renew Energy* 2022;182: 659–84.
- [12] Ramon J, Lledó L, Torralba V, Soret A, Doblas-Reyes FJ. What global reanalysis best represents near-surface winds? *Q J R Meteorol Soc* 2019;145(724):3236–51.
- [13] Jourdi B. Evaluation of ERA5, MERRA-2, COSMO-REA6, NEWA and AROME to simulate wind power production over France. *Adv Sci Res* 2020;17:63–77.
- [14] Gualtieri G. Reliability of ERA5 reanalysis data for wind resource assessment: a comparison against tall towers. *Energies* 2021;14(14):4169.
- [15] Colorado State University. Regional atmospheric modeling system (RAMS) model. 2021. <https://vandenheever.atmos.colostate.edu/vdhp/rams.php>. [Accessed 29 December 2021]. Accessed.
- [16] Chancham C, Waewsak J, Chaichana T, Landry M, Gagnon Y. Assessment of onshore wind energy potential using regional atmospheric modeling system (RAMS) for Thailand. *Energy Proc* 2014;52:487–96.
- [17] Pennsylvania State University/National Center for Atmospheric Research. MM5 model. 2021. <https://www2.mmm.ucar.edu/mm5/mm5v3.html>. [Accessed 29 December 2021]. Accessed.
- [18] Cheng FY, Georgakakos KP. Statistical analysis of observed and simulated hourly surface wind in the vicinity of the Panama Canal. *Int J Climatol* 2011;31(5): 770–82.
- [19] National Centre for Meteorological Research (CNRM). ALADIN model. 2021. <http://www.umr-cnrm.fr/aladin>. [Accessed 29 December 2021]. Accessed.
- [20] Horvath K, Bajić A, Ivatek-Šahdan S. Dynamical downscaling of wind resources in complex terrain of Croatia. In: *Proc. EWEC 2009*, marseille, France.
- [21] National Center for Atmospheric Research (NCAR). Weather research and forecasting (WRF) model. 2021. <https://www.mmm.ucar.edu/weather-research-and-forecasting-model>. [Accessed 29 December 2021]. Accessed.
- [22] Menendez MDCR, García-Díez M, Fita L, Fernández J, Méndez FJ, Gutiérrez JM. High-resolution sea wind hindcasts over the Mediterranean area. *Clim Dynam* 2014;42:1857–72.
- [23] Chancham C, Waewsak J, Gagnon Y. Offshore wind resource assessment and wind power plant optimization in the Gulf of Thailand. *Energy* 2017;139:706–31.
- [24] Carvalho D, Rocha A, Gómez-Gesteira M, Santos CS. Offshore wind energy resource simulation forced by different reanalyses: comparison with observed data in the Iberian Peninsula. *Appl Energy* 2014;134:57–64.
- [25] Niyomtham L, Lertsathittanakorn C, Waewsak J, Gagnon Y. On the wind resource assessment along the western coast of Thailand. *Energy Proc* 2017;138:1190–5.
- [26] Lorenz T, Barstad I. A dynamical downscaling of ERA-Interim in the North Sea using WRF with a 3 km grid—for wind resource applications. *Wind Energy* 2016; 19(10):1945–59.
- [27] Dörenkämper M, Olsen BT, Witha B, Hahmann AN, Davis NN, Barcons J, et al. The making of the new European wind atlas—part 2: production and evaluation. *Geosci Model Dev (GMD)* 2020;13: 5079–92.
- [28] Pickering B, Grams CM, Pfenninger S. Sub-national variability of wind power generation in complex terrain and its correlation with large-scale meteorology. *Environ Res Lett* 2020;15(4):044025.
- [29] Ruiz SAG, Barriga JEC, Martínez JA. Wind power assessment in the Caribbean region of Colombia, using ten-minute wind observations and ERA5 data. *Renew Energy* 2021;172:158–76.
- [30] Dodla VBR, Desamsetti S. Assessment of wind energy potential over India using high-resolution global reanalysis data. *J Earth Syst Sci* 2021;130(2):1–19.
- [31] Nefabas KL, Söder L, Mamo M, Olsson J. Modeling of Ethiopian wind power production using ERA5 reanalysis data. *Energies* 2021;14(9):2573.
- [32] Mahmoodi K, Ghassemi H, Razminia A. Wind energy potential assessment in the Persian Gulf: a spatial and temporal analysis. *Ocean Eng* 2020;216:107674.
- [33] Kumar VS, Asok AB, George J, Amrutha MM. Regional study of changes in wind power in the Indian shelf seas over the last 40 years. *Energies* 2020;13:2295.
- [34] Mattar C, Guzmán-Ibarra MC. A techno-economic assessment of offshore wind energy in Chile. *Energy* 2017;133:191–205.
- [35] Raileanu A, Onea F, Rusu E. Assessment of the wind energy potential in the coastal environment of two enclosed seas. In: *Oceans. Genova: IEEE*; 2015.
- [36] Draxl C, Mayr GJ. Meteorological wind energy potential in the Alps using ERA40 and wind measurement sites in the Tyrolean Alps. *Wind Energy* 2011;14(4): 471–89.
- [37] Olsson J. ERA5: the new champion of wind power modelling. *Renew Energy* 2018;126:322–31.
- [38] Miao H, Dong D, Huang G, Hu K, Tian Q, Gong Y. Evaluation of Northern Hemisphere surface wind speed and wind power density in multiple reanalysis datasets. *Energy* 2020;200:117382.
- [39] Fan W, Liu Y, Chappell A, Dong L, Xu R, Ekström M, et al. Evaluation of global reanalysis land surface wind speed trends to support wind energy development using in situ observations. *J Appl Meteorol Climatol* 2021;60(1):33–50.
- [40] Gruber K, Regner P, Wehrle S, Zeyringer M, Schmidt J. Towards global validation of wind power simulations: a multi-country assessment of wind power simulation from MERRA-2 and ERA-5 reanalyses bias-corrected with the global wind atlas. *Energy* 2022;238:121520.
- [41] Stopa JE, Cheung KF. Intercomparison of wind and wave data from the ECMWF reanalysis interim and the NCEP climate forecast system reanalysis. *Ocean Model* 2014;75:65–83.
- [42] Murcia JP, Koivisto MJ, Luzia G, Olsen BT, Hahmann AN, Sørensen PE, Als M. Validation of European-scale simulated wind speed and wind generation time series. *Appl Energy* 2022;305:117794.
- [43] Kaspar F, Niermann D, Borsche M, Fiedler S, Keller J, Potthast R, et al. Regional atmospheric reanalysis activities at Deutscher Wetterdienst: review of evaluation results and application examples with a focus on renewable energy. *Adv Sci Res* 2020;17:115–28.
- [44] Hussain M, Park SK. Systematic analysis of wind resources for eolic potential in Bangladesh. *Appl Sci* 2021;11(17):7924.
- [45] Sheridan LM, Krishnamurthy R, Gorton AM, Shaw WJ, Newsom RK. Validation of reanalysis-based offshore wind resource characterization using lidar buoy observations. *Mar Technol Soc J* 2020;54(6):44–61.
- [46] Yu J, Zhou T, Jiang Z, Zou L. Evaluation of near-surface wind speed changes during 1979 to 2011 over China based on five reanalysis datasets. *Atmosphere* 2019;10(12):804.
- [47] Kalnay E, Kanamitsu M, Kistler R, Collins W, Deaven D, Gandin L, et al. The NCEP/NCAR 40-year reanalysis project. *Bull Am Meteorol Soc* 1996;77:437–72.
- [48] Kanamitsu M, Ebisuzaki W, Woollen J, Yang SK, Hnilo JJ, Fiorino M, Potter GL. NCEP-DOE AMIP-II reanalysis (R-2). *Bull Am Meteorol Soc* 2002;83:1631–43.
- [49] Saha S, Moorthi S, Pan HL, Wu X, Wang J, Nadiga S, et al. The NCEP climate forecast system reanalysis. *Bull Am Meteorol Soc* 2010;91:1015–58.
- [50] Saha S, Moorthi S, Wu X, Wang J, Nadiga S, Tripp P, et al. The NCEP climate forecast system version 2. *J Clim* 2014;27:2185–208.
- [51] Rienecker MM, Suarez MJ, Gelaro R, Todling R, Bacmeister J, Liu E, et al. MERRA: NASA's modern-era retrospective analysis for research and applications. *J Clim* 2011;24:3624–48.
- [52] Gelaro R, McCarty W, Suárez MJ, Todling R, Molod A, Takacs L, et al. The Modern-Era retrospective analysis for research and applications, version 2 (MERRA-2). *J Clim* 2017;30:5419–54.
- [53] Uppala SM, Kallberg PW, Simmons AJ, Andrae U, Bechtold VD, Fiorino M, et al. The ERA-40 re-analysis. *Q J R Meteorol Soc* 2005;131:2961–3012.
- [54] Dee DP, Uppala SM, Simmons AJ, Berrisford P, Poli P, Kobayashi S, et al. The ERA-Interim reanalysis: configuration and performance of the data assimilation system. *Q J R Meteorol Soc* 2011;137:553–97.
- [55] Hersbach H, Bell B, Berrisford P, Hirahara S, Horányi A, Muñoz-Sabater J, et al. The ERA5 global reanalysis. *Q J R Meteorol Soc* 2020;146:1999–2049.
- [56] Kobayashi S, Ota Y, Harada Y, Ebata A, Moriya M, Onoda H, et al. The JRA-55 reanalysis: general specifications and basic characteristics. *J Meteorol Soc Jpn* 2015;93(Ser. II):5–48.
- [57] Mesinger F, DiMego G, Kalnay E, Mitchell K, Shafran PC, Ebisuzaki W, et al. North American regional reanalysis. *Bull Am Meteorol Soc* 2006;87(3):343–60.
- [58] Bollmeyer C, Keller JD, Ohlwein C, Wahl S, Crewell S, Friederichs P, et al. Towards a high-resolution regional reanalysis for the European CORDEX domain. *Q J R Meteorol Soc* 2015;141:1–15.
- [59] Wahl S, Bollmeyer C, Crewell S, Figura C, Friederichs P, Hense A, et al. A novel convective-scale regional reanalyses COSMO-REA2: improving the representation of precipitation. *Meteorol Z* 2017;26:345–61.
- [60] Ridal M, Olsson E, Undén P, Zimmermann K, Ohlsson A. HARMONIE reanalysis report of results and dataset. C, <https://www.ueru.eu/component/dpattachment/s/?task=attachment.download&id=297>. [Accessed 16 December 2021].
- [61] Jerney P, Davie J, Renshaw R. Ensemble variational DA documentation. UERRA Deliverable D2.4, <https://www.ueru.eu/component/dpattachments/?task=attachment.download&id=284>. [Accessed 16 December 2021].
- [62] Gleeson E, Whelan E, Hanley J. Met Éireann high resolution reanalysis for Ireland. *Adv Sci Res* 2017;14:49–61.
- [63] Bromwich DH, Wilson B, Bai L, Liu Z, Barlage M, Shih CF, et al. The Arctic system reanalysis, version 2. *B. Am Meteorol Soc* 2018;99:805–28.
- [64] Su CH, Eizenberg N, Steinle P, Jakob D, Fox-Hughes P, White CJ, et al. BARRA v1.0: the Bureau of meteorology atmospheric high-resolution regional reanalysis for Australia. *Geosci Model Dev (GMD)* 2019;12:2049–68.
- [65] Gruber K, Klöckl C, Regner P, Baumgartner J, Schmidt J. Assessing the Global Wind Atlas and local measurements for bias correction of wind power generation simulated from MERRA-2 in Brazil. *Energy* 2019;189:116212.
- [66] Core Team R. The R project for statistical computing. 2021. <https://www.r-project.org/>. [Accessed 19 December 2021].
- [67] R Graphics Package v. 4.2.0. Accessed 19/12/2021, <https://stat.ethz.ch/R-manual/R-devel/library/graphics/html/00Index.html>; 2021.
- [68] car R. Companion to applied regression package v. 3.0-12. 2021. 19/12/2021, <https://cran.r-project.org/web/packages/car/index.html>.
- [69] R boot package v. 1.3-28. 31/03/2022, <https://cran.r-project.org/web/packages/boot/boot.pdf>; 2021.
- [70] Hayes L, Stocks M, Blakers A. Accurate long-term power generation model for offshore wind farms in Europe using ERA5 reanalysis. *Energy* 2021;229:120603.
- [71] Duddy Clarke E, Griffin S, McDermott F, Monteiro Correia J, Sweeney C. Which reanalysis dataset should we use for renewable energy analysis in Ireland? *Atmosphere* 2021;12(5):624.
- [72] Borvarán D, Peña A, Gandoin R. Characterization of offshore vertical wind shear conditions in Southern New England. *Wind Energy* 2021;24(5):465–80.

- [73] Paiva V, Kampel M, Camayo R. Comparison of multiple surface ocean wind products with buoy data over blue amazon (Brazilian continental margin). *Adv Meteorol* 2021;6680626.
- [74] Brune S, Keller JD, Wahl S. Evaluation of wind speed estimates in reanalyses for wind energy applications. *Adv Sci Res* 2021;18:115–26.
- [75] Girleanu A, Onea F, Rusu E. Assessment of the wind energy potential along the Romanian coastal zone. *Inventions* 2021;6(2):41.
- [76] Livingston HG, Lundquist JK. How many offshore wind turbines does New England need? *Meteorol Appl* 2020;27(6):1969.
- [77] Morales-Ruvalcaba CF, Rodríguez-Hernández O, Martínez-Alvarado O, Drew DR, Ramos E. Estimating wind speed and capacity factors in Mexico using reanalysis data. *Energy Sustain Develop* 2020;58:158–66.
- [78] Hallgren C, Arnqvist J, Ivanell S, Körnich H, Vakkari V, Sahlée E. Looking for an offshore low-level jet champion among recent reanalyses: a tight race over the Baltic Sea. *Energies* 2020;13(14):3670.
- [79] Islek F, Yuksel Y, Sahin C. Spatiotemporal long-term trends of extreme wind characteristics over the Black Sea. *Dynam Atmos Oceans* 2020;90:101132.
- [80] De Assis Tavares LF, Shadman M, De Freitas Assad LP, Silva C, Landau L, Estefen SF. Assessment of the offshore wind technical potential for the Brazilian Southeast and South regions. *Energy* 2020;196:117097.
- [81] Frank CW, Pospichal B, Wahl S, Keller JD, Hense A, Crewell S. The added value of high resolution regional reanalyses for wind power applications. *Renew Energy* 2020;148:1094–109.
- [82] Ren G, Wan J, Liu J, Yu D. Characterization of wind resource in China from a new perspective. *Energy* 2019;167:994–1010.
- [83] Kaiser-Weiss AK, Borsche M, Niermann D, Kaspar F, Lussana C, Isotta FA, et al. Added value of regional reanalyses for climatological applications. *Environ Res Commun* 2019;1(7):071004.
- [84] Camargo LR, Gruber K, Nitsch F. Assessing variables of regional reanalysis data sets relevant for modelling small-scale renewable energy systems. *Renew Energy* 2019;133:1468–78.
- [85] Piasecki A, Jurasz J, Kies A. Measurements and reanalysis data on wind speed and solar irradiation from energy generation perspectives at several locations in Poland. *SN Appl Sci* 2019;1(8):1–8.
- [86] Albani A, Ibrahim MZ, Yong KH. Influence of the ENSO and monsoonal season on long-term wind energy potential in Malaysia. *Energies* 2018;11(11):2965.
- [87] Guozden TM, Bianchi E, Solarte A, Mulleady C. Wind resource assessment in the Río Negro province (Patagonia Argentina) using MERRA reanalysis. *Meteorologica* 2018;43(2):47–61.
- [88] Carvalho D, Rocha A, Gómez-Gesteira M, Santos CS. Offshore winds and wind energy production estimates derived from ASCAT, OSCAT, numerical weather prediction models and buoys—A comparative study for the Iberian Peninsula Atlantic coast. *Renew Energy* 2017;102:433–44.
- [89] Cradden LC, McDermott F, Zubiate L, Sweeney C, O'Malley M. A 34-year simulation of wind generation potential for Ireland and the impact of large-scale atmospheric pressure patterns. *Renew Energy* 2017;106:165–76.
- [90] Aniskevich S, Bezrukovs V, Zandovskis U, Bezrukovs D. Modelling the spatial distribution of wind energy resources in Latvia. *Latv J Phys Tech Sci* 2017;54(6):10.
- [91] Ritter M, Shen Z, Cabrera BL, Odening M, Deckert L. Designing an index for assessing wind energy potential. *Renew Energy* 2015;83:416–24.
- [92] Thomas N, Seim H, Haines S. An observational, spatially explicit, stability-based estimate of the wind resource off the shore of North Carolina. *J Appl Meteorol Climatol* 2015;54(12):2407–25.
- [93] Shar E, Dodds P, Barrett M, Spataru C. Evaluating the accuracy of CFSR reanalysis hourly wind speed forecasts for the UK, using in situ measurements and geographical information. *Renew Energy* 2015;77:527–38.
- [94] Carvalho D, Rocha A, Gómez-Gesteira M, Santos CS. Comparison of reanalyzed, analyzed, satellite-retrieved and NWP modelled winds with buoy data along the Iberian Peninsula coast. *Remote Sens Environ* 2014;152:480–92.
- [95] Pilão R, Pinto P, Guedes R, Carvalho L. Inter-annual wind speed variability over Polish territory. *EWEC: European Wind Energy Conference and Exhibition*; 2013.
- [96] Chawla A, Spindler DM, Tolman HL. Validation of a thirty year wave hindcast using the Climate Forecast System Reanalysis winds. *Ocean Model* 2013;70:189–206.
- [97] Liléo S, Petrik O. Investigation on the use of NCEP/NCAR, MERRA and NCEP/CFSR reanalysis data in wind resource analysis. *Sigma* 2000;1(2).
- [98] Kiss P, Varga L, János IM. Comparison of wind power estimates from the ECMWF reanalyses with direct turbine measurements. *J Renew Sustain Energy* 2009;1(3):033105.
- [99] Winterfeldt J, Andersson A, Klepp C, Bakan S, Weisse R. Comparison of HOAPS, QuikSCAT, and buoy wind speed in the eastern North Atlantic and the north sea. *IEEE Trans Geosci Rem Sens* 2009;48(1):338–48.
- [100] McQueen D, Watson S. Validation of wind speed prediction methods at offshore sites. *Wind Energy: Int J Progr Appl Wind Power Convers Technol* 2006;9(1–2):75–85.
- [101] Caires S, Sterl A, Bidlot JR, Graham N, Swail V. Intercomparison of different wind–wave reanalyses. *J Clim* 2004;17(10):1893–913.
- [102] Monforti F, Gonzalez-Aparicio I. Comparing the impact of uncertainties on technical and meteorological parameters in wind power time series modelling in the European Union. *Appl Energy* 2017;206:439–50.
- [103] Gonzalez-Aparicio I, Monforti F, Volker P, Zucker A, Careri F, Huld T, Badger J. Simulating European wind power generation applying statistical downscaling to reanalysis data. *Appl Energy* 2017;199:155–68.
- [104] Staffell I, Green R. How does wind farm performance decline with age? *Renew Energy* 2014;66:775–86.
- [105] Kubik M, Brayshaw D, Coker P, Barlow J. Exploring the role of reanalysis data in simulating regional wind generation variability over Northern Ireland. *Renew Energy* 2013;57:558–61.
- [106] Bosch J, Staffell I, Hawkes AD. Temporally explicit and spatially resolved global offshore wind energy potentials. *Energy* 2018;163:766–81.
- [107] Rose S, Apt J. What can reanalysis data tell us about wind power? *Renew Energy* 2015;83:963–9.
- [108] Technical University of Denmark (DTU). Global wind atlas (GWA) v. 3.1. <https://globalwindatlas.info>. [Accessed 29 December 2021].
- [109] Technical University of Denmark (DTU). WaSP 2021. <https://www.wasp.dk/WAS> P. [Accessed 29 December 2021].
- [110] Yu L, Zhong S, Bian X, Heilman WE. Climatology and trend of wind power resources in China and its surrounding regions: a revisit using Climate Forecast System Reanalysis data. *Int J Climatol* 2016;36(5):2173–88.
- [111] Boudia SM, Santos JA. Assessment of large-scale wind resource features in Algeria. *Energy* 2019;189:116299.
- [112] Tabata Y, Hashiguchi H, Yamamoto MK, Yamamoto M, Yamanaka MD, Mori S, et al. Lower tropospheric horizontal wind over Indonesia: a comparison of wind profiler network observations with global reanalyses. *J Atmos Sol Terr Phys* 2011;73:986–95.
- [113] Carvalho D, Rocha A, Gómez-Gesteira M, Santos CS. WRF wind simulation and wind energy production estimates forced by different reanalyses: comparison with observed data for Portugal. *Appl Energy* 2014;117:116–26.
- [114] Ingleby B. Global assimilation of air temperature, humidity, wind and pressure from surface stations. *Q J R Meteorol Soc* 2015;141:504–17.
- [115] Gholami S, Ghader S, Khaleghi-Zavareh H, Ghafari P. Sensitivity of WRF-simulated 10 m wind over the Persian Gulf to different boundary conditions and PBL parameterization schemes. *Atmos Res* 2021;247:105147.
- [116] Solbakken K, Birkelund Y, Samuelsen EM. Evaluation of surface wind using WRF in complex terrain: atmospheric input data and grid spacing. *Environ Model Software* 2021;145:105182.
- [117] Floors R, Nielsen M. Estimating air density using observations and Re-analysis outputs for wind energy purposes. *Energies* 2019;12:2038.
- [118] Aydoğan B. Offshore wind power atlas of the Black Sea region. *J Renew Sustain Energy* 2017;9:013305.
- [119] Ibarra-Berastegui G, Ulazia A, Sáenz J, González-Rojí SJ. Evaluation of Lebanon's offshore-wind-energy potential. *J Mar Sci Eng* 2019;7:361.
- [120] Ulazia A, Nafarrate A, Ibarra-Berastegui G, Sáenz J, Carreno-Madinabeitia S. The consequences of air density variations over Northeastern Scotland for offshore wind energy potential. *Energies* 2019;12:2635.
- [121] Ulazia A, Sáenz J, Ibarra-Berastegui G, González-Rojí SJ, Carreno-Madinabeitia S. Global estimations of wind energy potential considering seasonal air density changes. *Energy* 2019;187:115938.
- [122] Kanwal A, Tahir ZUR, Asim M, Hayat N, Farooq M, Abdullah M, et al. Evaluation of reanalysis and analysis datasets against measured wind data for wind resource assessment. *Energy Environ* 2022. <https://doi.org/10.1177/0958305X221084078>.
- [123] de Assis Tavares LF, Shadman M, de Freitas Assad LP, Estefen SF. Influence of the WRF model and atmospheric reanalysis on the offshore wind resource potential and cost estimation: a case study for Rio de Janeiro State. *Energy* 2022;240:122767.
- [124] Potisomporn P, Vogel CR. Spatial and temporal variability characteristics of offshore wind energy in the United Kingdom. *Wind Energy* 2021. <https://doi.org/10.1002/we.2685>.
- [125] Ahmad S, Abdullah M, Kanwal A, Tahir ZUR, Saeed UB, Manzoor F, et al. Offshore wind resource assessment using reanalysis data. *Wind Eng* 2022. <https://doi.org/10.1177/0309524X211069384>.
- [126] Pronk V, Bodini N, Optis M, Lundquist JK, Moriarty P, Draxl C, et al. Can reanalysis products outperform mesoscale numerical weather prediction models in modeling the wind resource in simple terrain? *Wind Energy Sci* 2022;7:487–504.
- [127] Hadjipetrou S, Liodakis S, Sykioti A, Katikas L, Park NW, Kalogirou S, et al. Evaluating the suitability of Sentinel-1 SAR data for offshore wind resource assessment around Cyprus. *Renew Energy* 2022;182:1228–39.
- [128] Sawadogo W, Reboita MS, Faye A, da Rocha RP, Odoulami RC, Olusegun CF, et al. Current and future potential of solar and wind energy over Africa using the RegCM4 CORDEX-CORE ensemble. *Clim Dynam* 2021;2021(57):1647–72.